

SEPSOR

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SEPSOR

Design History Document Concurrence

The undersigned acknowledge that they have reviewed and approve the SEPSOR Design History Document. Changes to this document must be coordinated and approved by the undersigned of their designated representatives.

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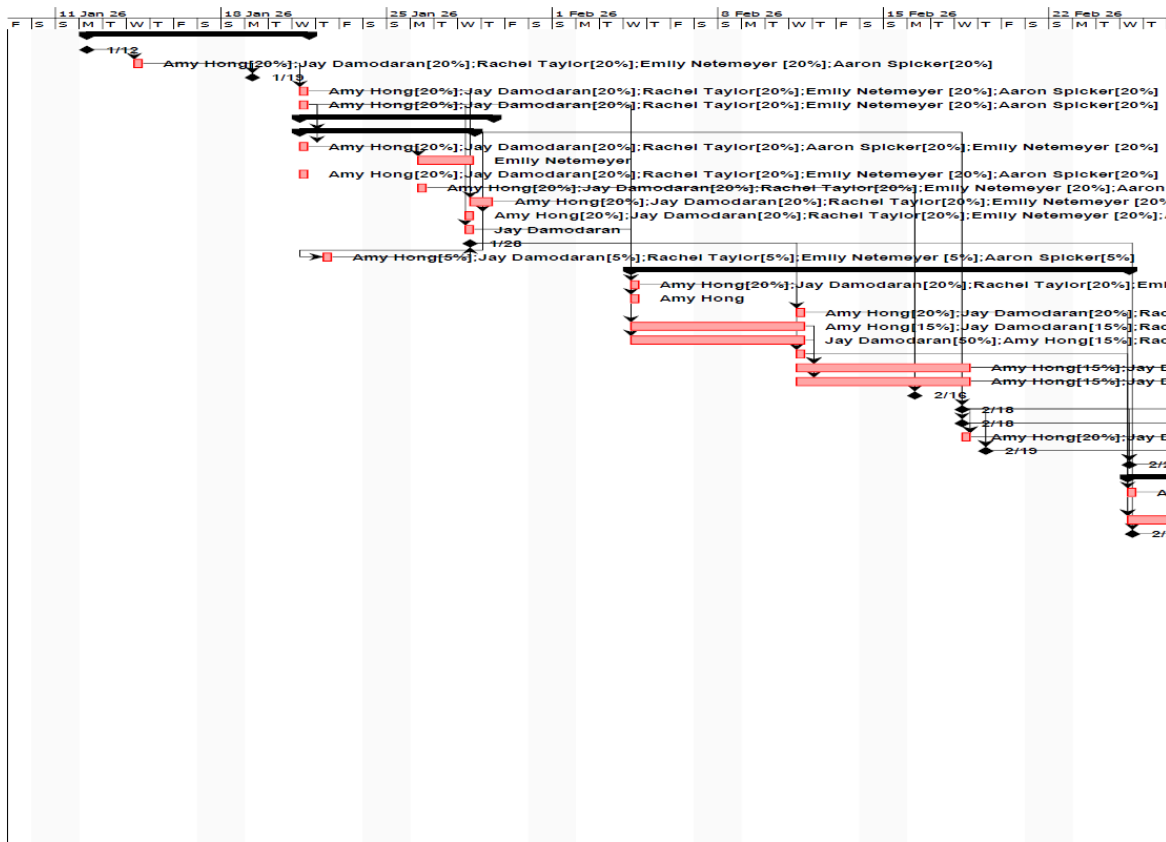
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Project Plan

		Name	Duration	Start	Finish	Predecessors	Resource Names
1		Idea Generation	9 days	1/12/26, 9:00 AM	1/21/26, 5:00 PM		
2		Class Start	0 days	1/12/26, 9:00 AM	1/12/26, 9:00 AM		
3		Week 1 Lab Work	1 day	1/14/26, 9:00 AM	1/14/26, 5:00 PM	2	Amy Hong[20%];Jay Demoda...
4		My Learning Outcomes	0 days	1/19/26, 9:00 AM	1/19/26, 9:00 AM	3	Amy Hong[20%];Jay Demoda...
5		Week 2 Lab Work	1 day	1/21/26, 9:00 AM	1/21/26, 5:00 PM	3	Amy Hong[20%];Jay Demoda...
6		Idea Finalized	1 day	1/21/26, 9:00 AM	1/21/26, 5:00 PM		Amy Hong[20%];Jay Demoda...
7		Research/Preparation	6.8 days	1/21/26, 9:00 AM	1/29/26, 1:00 PM		
8		Market Research/Custo...	6 days	1/21/26, 9:00 AM	1/29/26, 9:00 PM	6	
9		Collect Information	1 day	1/21/26, 9:00 AM	1/21/26, 5:00 PM	6	Amy Hong[20%];Jay Demoda...
10		Finalize/Summarize Infor...	3 days	1/26/26, 9:00 AM	1/29/26, 5:00 PM	9	Emily Natarayer
11		Teamwork Assessment	1 day	1/21/26, 9:00 AM	1/21/26, 5:00 PM		Amy Hong[20%];Jay Demoda...
12		Week 3 Biotech Industry D...	1 day	1/26/26, 9:00 AM	1/26/26, 5:00 PM		Amy Hong[20%];Jay Demoda...
13		Week 3 Lab Work	1 day	1/28/26, 12:00 PM	1/29/26, 1:00 PM	5	Amy Hong[20%];Jay Demoda...
14		Week 3 Teamwork Asses...	1 day	1/28/26, 9:00 AM	1/28/26, 5:00 PM	8	Amy Hong[20%];Jay Demoda...
15		Make Github	1 day	1/28/26, 9:00 AM	1/28/26, 5:00 PM	6	Jay Demodaran
16		Buying Supplies	0 days	1/28/26, 12:00 PM	1/28/26, 12:00 PM	17	Amy Hong[5%];Jay Demoda...
17		Create Finance Sheet	1 day	1/22/26, 9:00 AM	1/22/26, 5:00 PM	8	Amy Hong[5%];Jay Demoda...
18		Planning/Prototyping	15.128 days	2/4/26, 9:00 AM	2/25/26, 9:00 AM		
19		Week 4 Lab Work	1 day	2/4/26, 9:00 AM	2/4/26, 5:00 PM	13	Amy Hong[20%];Jay Demoda...
20		Project Schedule	1 day	2/4/26, 9:00 AM	2/4/26, 5:00 PM	6	Amy Hong
21		Week 5 Lab Work	1 day	2/11/26, 9:00 AM	2/11/26, 5:00 PM	19	Amy Hong[20%];Jay Demoda...
22		Start Prototype Design	6 days	2/4/26, 9:00 AM	2/11/26, 5:00 PM	16	Amy Hong[15%];Jay Demoda...
23		Start Preliminary Coding	6 days	2/4/26, 9:00 AM	2/11/26, 5:00 PM	15	Jay Demodaran[50%];Amy Ho...
24		Parts Arrive	1 day	2/11/26, 9:00 AM	2/11/26, 5:00 PM	16	
25		Design Caga	6 days	2/11/26, 9:00 AM	2/18/26, 5:00 PM	22;24	Amy Hong[15%];Jay Demoda...
26		Finalize Coding	6 days	2/11/26, 9:00 AM	2/18/26, 5:00 PM	23	Amy Hong[15%];Jay Demoda...
27		Week 6 Biotech Industry Dis...	0 days	2/18/26, 9:00 AM	2/18/26, 9:00 AM	12	Amy Hong[20%];Jay Demoda...
28		Requirements Specification...	0 days	2/18/26, 9:00 AM	2/18/26, 9:00 AM	8	Amy Hong[20%];Jay Demoda...
29		Requirements Traceability Ma...	0 days	2/18/26, 9:00 AM	2/18/26, 9:00 AM	8	Amy Hong[20%];Jay Demoda...
30		Week 6 Teamwork Asses...	1 day	2/18/26, 9:00 AM	2/18/26, 5:00 PM	28;29	Amy Hong[20%];Jay Demoda...
31		Requirements Review Minu...	0 days	2/19/26, 9:00 AM	2/19/26, 9:00 AM	28;29	Amy Hong[10%];Jay Demoda...
32		Requirements Specification...	0 days	2/25/26, 9:00 AM	2/25/26, 9:00 AM	28;29	Amy Hong[20%];Jay Demoda...
33		Assembling/Presentation	44 days	3/25/26, 9:00 AM	4/27/26, 9:00 PM		
34		Week 7 Lab Work	1 day	2/25/26, 9:00 AM	2/25/26, 5:00 PM	21	Amy Hong[20%];Jay Demoda...
35		Week 8 Lab Work	1 day	3/4/26, 9:00 AM	3/4/26, 5:00 PM	34	Amy Hong[20%];Jay Demoda...
36		Design PCB	6 days	2/25/26, 9:00 AM	3/4/26, 5:00 PM	24;26	Amy Hong[15%];Jay Demoda...
37		Start Assembling Product	0 days	2/25/26, 12:00 PM	2/25/26, 12:00 PM	16	Amy Hong[20%];Jay Demoda...
38		Start 3D Printing	1 day	3/2/26, 9:00 AM	3/2/26, 5:00 PM	25;26	Amy Hong[20%];Jay Demoda...
39		Week 9 Biotech Industry D...	1 day	3/9/26, 9:00 AM	3/9/26, 5:00 PM	30	Amy Hong[20%];Jay Demoda...
40		BHE Job Requisition	1 day	3/9/26, 9:00 AM	3/9/26, 5:00 PM		Amy Hong[20%];Jay Demoda...
41		Week 9 Lab Work	1 day	3/11/26, 9:00 AM	3/11/26, 5:00 PM	35	Amy Hong[20%];Jay Demoda...
42		Week 9 Teamwork Asses...	1 day	3/11/26, 9:00 AM	3/11/26, 5:00 PM		Amy Hong[20%];Jay Demoda...
43		Complete Caga	1 day	3/11/26, 9:00 AM	3/11/26, 5:00 PM	25	Amy Hong[15%];Jay Demoda...
44		Week 10 Lab Work	1 day	3/25/26, 9:00 AM	3/25/26, 5:00 PM	43	Amy Hong[20%];Jay Demoda...
45		Order Custom PCBs	0 days	3/25/26, 9:00 AM	3/25/26, 9:00 AM	28;43	Amy Hong[10%];Jay Demoda...
46		Start Presentation	4 days	3/25/26, 9:00 AM	3/30/26, 5:00 PM	38;43;45	Amy Hong[45%];Jay Demoda...
47		Finalize Product	0 days	3/31/26, 9:00 AM	3/31/26, 9:00 AM	37;38;43	Amy Hong[20%];Jay Demoda...
48		Design Specifications Docu...	0 days	4/1/26, 9:00 AM	4/1/26, 9:00 AM	28;29;31;32	Amy Hong[15%];Jay Demoda...
49		Requirements Traceability ...	0 days	4/1/26, 9:00 AM	4/1/26, 9:00 AM	28;29;31;32	Amy Hong[15%];Jay Demoda...
50		Design Review Minutiae	0 days	4/2/26, 9:00 AM	4/2/26, 9:00 AM	48;49	Amy Hong[10%];Jay Demoda...
51		Week 12 Biotech Industry ...	1 day	4/6/26, 9:00 AM	4/6/26, 5:00 PM	39	Amy Hong[20%];Jay Demoda...
52		Week 12 Lab Work	1 day	4/8/26, 9:00 AM	4/8/26, 5:00 PM	44	Amy Hong[20%];Jay Demoda...
53		Week 12 Teamwork Asses...	1 day	4/8/26, 9:00 AM	4/8/26, 5:00 PM	42	Amy Hong[20%];Jay Demoda...
54		Week 13 Code Review	0 days	4/15/26, 9:00 AM	4/15/26, 9:00 AM	47	Amy Hong[15%];Jay Demoda...
55		Week 13 Lab Work	1 day	4/15/26, 9:00 AM	4/15/26, 5:00 PM	52	Amy Hong[20%];Jay Demoda...
56		Finalize Presentation	0 days	4/15/26, 9:00 AM	4/15/26, 9:00 AM	46;47	Amy Hong[60%];Jay Demoda...
57		Design History File	0 days	4/23/26, 9:00 AM	4/23/26, 9:00 AM		Amy Hong[20%];Jay Demoda...
58		Final Design Oral Presentat...	0 days	4/23/26, 9:00 AM	4/23/26, 9:00 AM	47;56	Amy Hong[50%];Jay Demoda...
59		Reflection Essay	1 day	4/27/26, 9:00 AM	4/27/26, 5:00 PM		Amy Hong[15%];Jay Demoda...





Statement of Purpose, Customers, and Device Use

This device is designed to allow for the early detection of sepsis via the monitoring of three key physiological markers: heart rate (HR) and respiratory rate (RR). While sepsis is capable of being monitored or diagnosed in clinical settings, environments without direct medical access can cause sepsis to go unnoticed. Therefore, our primary customers are individuals at risk of developing sepsis and are outside of continuous hospital care including recently discharged patients, post-operative patients, those in outpatient programs, and other high-risk individuals such as those with underlying conditions or injuries.

The primary purpose of this device is to alert the user to physiological indicators of sepsis to reduce the time-to-care of an otherwise symptom-based condition, as early detection greatly improves survival and health outcomes. The design of this device allows for continuous and noninvasive monitoring throughout day-to-day life, providing simple and affordable peace of mind during home recovery. This use philosophy emphasizes a focus on reliability, accessibility, and ease of use to promote a near passive experience for users.

Customer Needs:

1. The device needs to be operational and produce accurate readings
2. The device should retain accurate function in reasonable conditions such as cold/heat/sweat as well as states of motion
3. This device should utilize its sensor readings to assist in sepsis detection within a reasonable timeframe (before I feel significantly ill)
4. This device should be non-invasive and entirely safe for the user.
5. This device should be easily utilized and adjusted without the need for assistance.
6. This device should be comfortable enough for long-term, multi-day use.
7. The battery of this device should allow for uninterrupted use throughout a typical day.
8. The battery of this device should be capable of a reasonable recharge speed.
9. The device should clearly communicate its warning when it is triggered.
10. This device should provide an understanding of what to do if an alert occurs.
11. This device should clearly communicate what it does.
12. This device should minimize false positive alerts.
13. This device should operate as a private, standalone system with relative independence from other connections or environments.
14. The device shall withstand normal daily activities.
15. The device shall be lightweight and discreet for daily wear.
16. The device shall be safe for continuous skin contact.

Requirements Traceability Matrix

Requirements Traceability Matrix			
Project Name	Sepsor		
Last update	4/1/2026		
ID	Customer need	Customer need document	
1	The device should be easy to put on, adjust, and operate without assistance.	5	
2	The device needs to be operational and continuously provide physiological monitoring data.	1	
3	The device needs to be operational and continuously provide physiological monitoring data.	1	
4	The device should be easy to put on, adjust, and operate without assistance.		
5	The battery of this device should be capable of a reasonable recharge speed.	5, 8	
6	The device should be easy to put on, adjust, and operate without assistance.	5	
7	This device should be comfortable enough for long-term, multi-day use.		
8	The device should be lightweight and discreet for daily wear.	6, 15	
9	The device should withstand normal daily activities.	14	
10	The device should be easy to put on, adjust, and operate without assistance.	5	
11	The device should withstand normal daily activities.	14	
	The device should be easy to put on, adjust, and operate without assistance.	5	
	The device should be non-invasive and worn externally on the body.		
12	This device should be comfortable enough for long-term, multi-day use.		
	The device should be lightweight and discreet for daily wear.		
13	The device should be safe for continuous skin contact.	4, 6, 15, 16	
14	The device needs to be operational and continuously provide physiological monitoring data.		
15	The device should provide early warning of worsening physiological condition.	1, 3	
	The device should provide early warning of worsening physiological condition.	3, 9	
16	The device should be non-invasive and worn externally on the body.	4	
17	The device needs to be operational and continuously provide physiological monitoring data.		
18	This device should minimize false positive alerts.	1, 12	
	The device needs to be operational and continuously provide physiological monitoring data.	1, 12	
	The device needs to be operational and continuously provide physiological monitoring data.		
19	This device should minimize false positive alerts.	1, 12	
	The device needs to be operational and continuously provide physiological monitoring data.		
20	The device should retain accurate function in reasonable conditions such as cold/heat/sweat as well as states of motion		
	This device should minimize false positive alerts.		
	The device should withstand normal daily activities.		
21		1, 2, 12, 14	
22	The device needs to be operational and continuously provide physiological monitoring data.		
23	The device should provide early warning of worsening physiological condition.		
24	This device should minimize false positive alerts.	1, 3, 12	
25	The device needs to be operational and continuously provide physiological monitoring data.		
	This device should minimize false positive alerts.	1, 12	
26			
27	The battery of this device should allow for uninterrupted use throughout a typical day.	7	
	The battery of this device should be capable of a reasonable recharge speed.	8	
28	The battery of this device should allow for uninterrupted use throughout a typical day.	7	
29	This device should be comfortable enough for long-term, multi-day use.		
	The device should be lightweight and discreet for daily wear.	6, 15	
30	This device should be comfortable enough for long-term, multi-day use.		
	The device should be lightweight and discreet for daily wear.	6, 15	
31	The device should retain accurate function in reasonable conditions such as cold/heat/sweat as well as states of motion	2	
32	The device should be non-invasive and worn externally on the body.		
	The device should be safe for continuous skin contact.	4, 16	
	The device should be non-invasive and worn externally on the body.		
	This device should operate as a private, standalone system with relative independence from other connections or environments.	4, 13	

33	The device should be non-invasive and worn externally on the body. The device should be safe for continuous skin contact.	4, 16
34	The device should be non-invasive and worn externally on the body. The device should be safe for continuous skin contact.	4, 16
35	The device should retain accurate function in reasonable conditions such as cold/heat/sweat as well as states of motion The device should withstand normal daily activities.	16
36	The device should clearly communicate its warning when it is triggered. This device should provide an understanding of what to do if an alert occurs. This device should clearly communicate what it does.	2, 14
37	The device should clearly communicate its warning when it is triggered. This device should provide an understanding of what to do if an alert occurs. This device should clearly communicate what it does.	9, 10, 11
	The device should clearly communicate its warning when it is triggered. This device should provide an understanding of what to do if an alert occurs. This device should clearly communicate what it does.	9, 10, 11
	The device should clearly communicate its warning when it is triggered. This device should provide an understanding of what to do if an alert occurs. This device should clearly communicate what it does.	9, 10, 11
	The device should clearly communicate its warning when it is triggered. This device should provide an understanding of what to do if an alert occurs. This device should clearly communicate what it does.	9, 10, 11
38	This device should operate as a private, standalone system with relative independence from other connections or environments.	13

Requirements	Design	Implementation	
Requirement specification document	Design specification document	Component	Software module
3.1.1.1	4.1.3.1-4.1.3.5		
3.1.1.3	4.1.1.1-4.1.1.7		
3.1.1.5	3.1.1.5-3.1.1.6		
3.1.1.4	4.1.4; 3.4.9		
3.1.1.5	4.1.3.1-4.1.3.5		
3.1.2.1	4.1.2.1-4.1.2.3		
3.1.2.2	4.1.2.3-4.1.2.4		
3.1.2.3	4.1.3.1-4.1.3.5		
3.1.2.4	4.1.3.1-4.1.3.5		
3.1.3.1	4.1.2.3		
3.1.3.2	4.1.3.1-4.1.3.5		
3.1.3.3	4.1.2.1-4.1.2.3; 4.1.3.2-4.1.3.7		
3.2.1.1	4.1.1.1-4.1.1.7		

3.2.1.4	3.2.1-3.2.4		
3.2.1.5	4.1.2; 4.1.3		
3.2.2.1	4.1.1.1-4.1.1.7		
3.2.2.2	4.1.1.1-4.1.1.7		
3.2.2.3	4.1.3.1-4.1.3.5		
3.2.2.4	4.1.2.3; 4.1.3.1-4.1.3.5		
3.2.2.5	4.1.1.1-4.1.1.7		
3.2.2.6	4.1.1.1-4.1.1.7; 4.1.3.1-4.1.3.5		
3.2.3.1	4.1.4		
3.2.3.2	4.1.4; 3.4.9		
3.2.3.3	4.1.4		
3.2.3.4	4.1.2.1-4.1.2.6		
3.2.3.5	4.1.2.1-4.1.2.3		
3.2.3.6	4.1.2.3		
3.2.4.1	4.1.3.2-4.1.3.7; 4.1.2.1		
3.2.4.2	3.1.1.6; 4.1.2		
3.2.4.3	4.1.3.2; 4.1.2.1		
3.2.4.4	4.1.3.2-4.1.3.3		
3.3.1.1	4.1.3.2; 4.1.2.1		
3.3.1.2	4.1.2.3; 4.1.3.4-4.1.3.5		
3.3.2.1	3.2.1-3.2.4		
3.3.2.2	3.4.12		
3.3.2.3	3.2.3		
3.3.2.4	3.4.12		

3.3.3.1	3.1.1.1–3.1.1.6		
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Additional Comments
ease of use
accuracy and continuous monitoring
accuracy and continuous monitoring
ease of use, recharging convenience
ease of use
comfort and wearability
accuracy and durability
ease of use
ease of use
accuracy and durability
ease of use
safety and biocompatibility, comfort and wearability
accuracy and continuous monitoring, trend-based detection with timely alerts
trend-based detection with timely alerts, alerts and user notification
safety and biocompatibility
accuracy and continuous monitoring, accuracy and filtering
accuracy and continuous monitoring, accuracy and filtering
accuracy and continuous monitoring, accuracy and filtering
accuracy and continuous monitoring, environmental robustness, accuracy and filtering, accuracy and durability
accuracy and continuous monitoring, trend-based detection with timely alerts, accuracy and filtering
accuracy and continuous monitoring, accuracy and filtering
battery life
recharging convenience
battery life
comfort and wearability
comfort and wearability
environmental robustness
safety and biocompatibility
safety and biocompatibility, privacy

safety and biocompatibility
safety and biocompatibility
safety and biocompatibility
environmental robustness, accuracy and durability
alerts and user notification, user guidance, operational transparency
alerts and user notification, user guidance, operational transparency
alerts and user notification, user guidance, operational transparency
alerts and user notification, user guidance, operational transparency
privacy

Requirements Specifications Document

for

SEPSOR

Version: 1.1

Prepared by:

Aaron Spicker

Amy Hong

Emily Netemeyer

Jayanth Damodaran

Rachel Taylor

Last modified: 02/25/202

1.0 General Information

1.1 Purpose

The purpose of this Requirements Specifications Document is to define the functionality, safety, usability, and regulatory requirements for the proposed medical device designed for early detection of sepsis. This document outlines the user, system, performance, software, and interface requirements that will be used for the testing and validation of the device. It also acts as an authority on the implementation of customer needs, regulatory requirements, engineering constraints, and other expectations into clear, verifiable design elements.

Finally, it establishes assessment criteria regarding the device/system and its performance, safety, usability, and interfacing in order to provide a baseline for future steps such as design verification and validation, risk assessment, and requirements tracing. Henceforth, all design decisions are understood to remain consistent with and uphold these defined requirements throughout the process of development.

1.2 Scope

This document outlines the relevant design requirements of the SEPSOR physiological monitoring system designed for early detection of sepsis. It establishes assumptions of use and the user while governing requirements related to but not limited to device operation and wearability, monitoring parameters, sepsis risk determination, measurement accuracy, power capabilities, electrical safety, generation of alerts, and notification behaviors. While physiological monitoring as well as risks and assumptions are outlined, there are no official diagnostic or treatment capabilities present, and as such no medical care obligations or provider relationships are established nor fall under the scope of this document.

The contents of this document serve only to specify design minimum requirements by providing guidelines for planning and development and metrics for integration and verification testing. In addition to the elements not included in this document, process specifications, supplier details, clinical testing, and regulatory submissions, as well as all other downstream market activities are outside the scope of this document.

1.3 Referenced documents

1.3.1 Customer Needs Document

1.3.2 Requirements Traceability Matrix

1.3.3 Cleveland Clinic: Articles – Vital Signs

1.3.4 ASHRAE Humidity Control Resources

1.3.5 FDA 21 CFR 820

1.3.6 ISO 10993

1.3.7 IEC 60601-1

1.3.7.1 IEC 60601-1-2

1.3.7.2 IEC 60601-1-8

1.3.8 IEEE 1708

1.3.9 DHHS (NIOSH) Publication No. 98-126

1.3.10 IEC 60529

1.4 Acronyms and Abbreviations

1.4.1 HR: Heart Rate

1.4.2 RR: Respiratory Rate

1.4.3 CN: Customer Need

1.4.4 UR: User Requirements

1.4.5 mmHG: Millimeter of Mercury

1.4.6 BPM: Beats per minute

1.4.7 ms: Millisecond

1.4.8 mm: Millimeter

1.4.9 °F: Degrees Fahrenheit

1.4.10 RAM: Random Access Memory

1.4.11 ROM: Read Only Memory

2.0 Overall Description

This device is designed to allow for the early detection of sepsis via the monitoring of three key physiological markers: HR, RR, and BP. Sepsis is capable of being monitored or diagnosed in clinical settings, environments without direct medical access can cause sepsis to go unnoticed. Therefore, our primary customers are individuals at risk of developing sepsis and are outside of continuous hospital care including recently discharged patients, post-operative patients, those in outpatient programs, and other high-risk individuals such as those with underlying conditions or injuries.

The primary purpose of this device is to alert the user to physiological indicators of sepsis to reduce the time-to-care of an otherwise symptom-based condition, as early detection greatly improves survival and health outcomes. The design of this device allows for continuous and noninvasive monitoring throughout day-to-day life, providing simple and affordable peace of mind during home recovery. This use philosophy emphasizes a focus on reliability, accessibility, and ease of use to promote a near passive experience for users.

2.1 Assumptions

The following assumptions define the expected user population, operating environment, and system usage conditions. These assumptions are made to bound the scope of the design and reduce engineering complexity. They represent reasonable expectations for the intended use case; however, they may limit the overall inclusivity and addressable market of the device.

2.1.1 User Assumptions

- 2.1.1.1 The intended user is a patient identified by a licensed healthcare provider as being at risk for developing sepsis.
- 2.1.1.2 The intended user is eligible to receive physiological monitoring outside of an acute inpatient hospital setting.
- 2.1.1.3 The intended user has baseline HR values within 60 to 100 BPM.
- 2.1.1.4 The intended user has baseline RR values within 12 to 18 breaths per minute.
- 2.1.1.5 The device is intended for use during routine daily activities.
- 2.1.1.6 The device is intended for use during low intensity physical activities.
- 2.1.1.7 The intended user is able to hear audible alerts at the defined sound levels.
- 2.1.1.8 The user is cognitively able to recognize the specified alert sounds.
- 2.1.1.9 The user is able to respond to the specific alert sounds appropriately.
- 2.1.1.10 The intended user can independently turn on the device without assistance.
- 2.1.1.11 The intended user can recharge the device without assistance.
- 2.1.1.12 The intended user can reposition the device without assistance.
- 2.1.1.13 The intended user has intact skin at the wear location.
- 2.1.1.14 The intended user has no known allergies to medical-grade adhesive.

- 2.1.1.15 The intended user has no known skin sensitivity to medical-grade adhesive.
- 2.1.1.16 The intended user has no known allergies to the strap materials.
- 2.1.1.17 The intended user has no known skin sensitivity to the strap materials.
- 2.1.1.18 The intended user shall follow the defined charging schedule to maintain continuous monitoring capability.

2.1.2 Environmental Assumptions

- 2.1.2.1 The device is primarily used in indoor, temperature-controlled environments, representative of a home recovery setting.
- 2.1.2.2 The ambient operating temperature is between 60-80 °F.
- 2.1.2.3 The ambient operating humidity levels are between 30 to 60%.
- 2.1.2.4 The device is exposed to minimal amounts of water from sweat.
- 2.1.2.5 The device is exposed to minimal amounts of water from normal skin moisture.
- 2.1.2.6 Electromagnetic interference levels are consistent with typical residential environments.

2.1.2 Operational Assumptions

- 2.1.3.1 The device shall be used as an early-warning tool for sepsis.
- 2.1.3.2 The device shall be used as a screening tool rather than a standalone diagnostic system for sepsis.
- 2.1.3.3 Sensor-to-skin contact is always maintained during normal use to ensure valid physiological measurements.
- 2.2.3.4 The device is worn in the intended chest location
- 2.2.3.5 The device is worn in the intended orientation.

2.3 Constraints

- 2.3.1 The device must comply with IEC 60601-1 electrical safety requirements.
- 2.3.2 The device must comply with IEC 60601-1-2 electromagnetic compatibility requirements.
- 2.3.3 Audible alerts must comply with IEC 60601-1-8 alarm standards
- 2.3.4 All skin-contacting materials must meet ISO 10993 biocompatibility requirements.
- 2.3.4 Device components must cost no more than \$100

3.0 Requirements

3.1 User Requirements

3.1.1 Apparatus

- 3.1.1.1 The user shall be able to secure the device on the anterior chest without assistance within a total of 2 minutes using only the provided attachment method.
- 3.1.1.2 The device shall operate continuously while battery power is available without requiring user activation after initial placement.
- 3.1.1.3 The device shall automatically initiate physiological monitoring upon placement without user intervention.
- 3.1.1.4 The user shall be able to recharge the device using a standard USB-C cable.
- 3.1.1.5 The device shall operate during normal daily use without requiring routine configuration by the user.
- 3.1.1.6 The device shall operate during normal daily use without requiring routine calibration by the user.
- 3.1.1.7 The device shall operate during normal daily use without requiring routine adjustment of the device parameters by the user.

3.1.2 Strap

- 3.1.2.1 The strap shall be adjustable to fit chest circumferences ranging from 38 to 48 inches.
- 3.1.2.2 The strap shall maintain device position with a maximum displacement of 10 mm during normal daily movement.
- 3.1.2.3 The strap shall maintain device position with a maximum displacement of 10 mm during sleep.
- 3.1.2.4 The user shall be able to remove and reattach the strap without assistance within a total of 2 minutes.
- 3.1.2.4 The strap shall be adjustable without the use of tools.
- 3.1.2.6 The device shall be removable without the use of tools.
- 3.1.2.7 The device shall be reattached without the use of tools.

3.1.3 Adhesive

- 3.1.3.1 The adhesive interface shall maintain the device in a secure position for at least 24 hours during normal daily movement.

3.1.3.2 The adhesive interface shall maintain the device in a secure position for at least 24 hours during sleep.

3.1.3.3 The user shall be able to remove the adhesive without assistance.

3.1.3.4 The user shall be able to reapply the adhesive without assistance.

3.1.3.5 The adhesive shall demonstrate biocompatibility with the contacted skin during the intended duration.

3.2 System Requirements

3.2.1 Functional Performance

3.2.1.1 The device shall monitor HR for 3 minutes every 15 minutes while powered to provide trend-based physiological data.

3.2.1.4 The device shall compute a sepsis risk score at intervals of 15 minutes to ensure timely detection.

3.2.1.5 The device shall require 2 consecutive abnormal readings before generating a sepsis alert.

3.2.1.6 The device shall alert the user within 60 seconds of confirmation of abnormal physiological conditions.

3.2.1.7 The device shall perform all physiological measurements using non-invasive sensing methods.

3.2.2 Measurement & Accuracy

3.2.2.1 HR measurement shall achieve ± 10 bpm accuracy compared to a clinical reference device to ensure suitable early detection.

3.2.2.2 RR measurement shall achieve ± 5 breaths/minute accuracy compared to a clinical reference device to ensure suitable early detection.

3.2.2.3 Measurements shall maintain accuracy within the specified limits during normal daily movement to support real-world daily use.

3.2.2.4 Data streams shall be synchronized within ± 50 ms to ensure suitable early detection.

3.2.2.5 The device shall filter raw physiological data to produce usable signals to ensure clean and interpretable readings.

3.2.3 Power & Physical Characteristics

3.2.3.1 The device shall operate ≥ 12 hours between charges to ensure continuous monitoring without user intervention.

3.2.3.2 The battery shall recharge from 0–80% in ≤ 60 minutes to minimize monitoring downtime.

3.2.3.3 The device shall provide a low-battery warning at $\leq 20\%$ charge to give user sufficient time to recharge device.

3.2.3.4 The device shall weigh ≤ 400 grams to maintain comfort and wearability.

3.2.3.5 The device dimensions shall be ≤ 55 mm x 100 mm to ensure compatibility and comfort for the user.

3.2.3.6 The device shall meet minimum IPX4 ingress protection to ensure safe and reliable reading during daily activities.

3.2.4 Safety & Compliance

3.2.4.1 The device shall meet IEC 60601-1 electrical safety requirements to ensure safe operation for the user.

3.2.4.2 The device shall meet IEC 60601-1-2 EMC requirements to prevent electromagnetic interference.

3.2.4.3 Materials contacting skin shall meet ISO 10993 biocompatibility standards to minimize the risk of irritation.

3.2.4.4 The charging interface shall prevent unsafe operation during charging to eliminate risk associated with simultaneous charging.

3.2.5 Hardware Requirements

3.2.5.1 The device shall include a microcontroller unit.

3.2.5.2 The microcontroller shall include at least 1 hardware timer.

3.2.5.3 The microcontroller shall include at least 1 Pulse Width Modulation pin.

3.2.5.4 The microcontroller shall provide hardware support for digital serial communication protocols to interface with sensors.

3.2.5.5 The microcontroller shall include at least one analog input pin.

3.2.5.6 The microcontroller shall have at least 75 kB RAM. 3.2.5.7 The microcontroller shall have at least 70 kB ROM.

3.2.5.8 The device shall include a buzzer.

3.2.5.9 The device shall be powered by rechargeable Lipo Battery.

3.2.5.10 The device shall include a Lipo Charging Unit/Battery Management System for electrical safety.

3.2.5.11 The device shall have sensor(s) capable of measuring HR

3.2.5.12 The device shall have sensor(s) capable of measuring RR

3.2.6 Software Requirements

3.2.6.1 The device software shall run exclusively on the microcontroller unit.

3.2.6.2 The device shall sample from sensors at a minimum of 120Hz. 3.2.6.3 The device software shall perform digital signal processing on raw data.

3.2.6.4 The device software shall compute RR every 30 seconds during the active period.

3.2.6.5 The device software shall compute HR every 2 seconds during the active period.

3.2.6.6 The device software shall read the analog input pin connected to the battery once at the end of every active period.

3.2.6.7 The device software shall sound the buzzer to indicate low battery.

3.2.6.8 The device software shall sound the buzzer to indicate sepsis warning.

3.2.6.9 The device software shall alert the user of sepsis when all of the following conditions are met:

3.2.6.9.1 HR > 90 BPM.

3.2.6.9.2 RR > 22 breaths/minute.

3.2.6.10 The device software shall trigger state change to low-power mode after 3 minutes in active mode.

3.2.6.11 The device software shall trigger state change to active mode after 12 minutes in low power mode.

3.3 Interface Requirements

3.3.1 Device–Body Interface

3.3.1.1 The device shall interface with the body using medical-grade adhesive.

3.3.1.2 The device shall interface with the body via an adjustable strap.

3.3.1.2.1 The strap shall be tightened circumferentially around the thoracic region.

3.3.1.3 The interface shall maintain functional performance following exposure to liquid ingress conditions consistent with an IPX4 classification per IEC 60529

3.3.2 Device–User Alert Interface

3.3.2.1 The device shall provide audible alerts compliant with IEC 60601-1-8.

3.3.2.2 Alert signals shall have a sound pressure level of at least 70 dB at 1 meter.

3.3.2.3 Alert signals shall include a pulsed tone with a repetition period between 1.5 and 2 seconds.

3.3.2.4 Alert signals shall persist until acknowledged by the user.

3.3.2.3 Alert volume shall be perceivable in environments up to 60 dB.

3.3.2.4 The device shall provide a distinct audible pattern for sepsis alerts.

3.3.2.5 The device shall provide a distinct audible pattern to indicate low battery

Requirements Specification Minutes

General Information:

update acronyms and abbreviations (like ms or temperature)

add formatting to sections 1.3 and 1.4

- 2.0: update spelling out abbreviations that are stated above

Overall Description:

- No notes

Assumptions:

- Split up 2.1.1.6 into allergies and intact skin
- Split up each requirement that has conjunctions into different requirements
- 2.1.1.7 use shall instead of will
- 2.1.1.3 needs to update to not include “not”
- Indentations needed for format Constraints:
- Add budget

Requirements:

- Shorten 3.1.1.1?
- 3.1.3.3 does not make sense
- 3.2.2 needs updating on acronyms
- 3.2.3.5 needs depth
- 3.2.3.6 not needed since it is stated in general
- 3.2.3.7 needs updating on documentation
- 3.2.4.4 needs updating on wording
- 3.2.4.3 needs updating on material
- 3.3.1.1 needs updating on material
- 3.3.2.2 needs to be separated
- 3.3.2.4 needs to be separated
- 3.3.1.1 needs to be separated
- Delete 3.3.3.1
- More information needed on sensors ordered
- Cite measurement and accuracy

Detailed Design Specifications Document

for

SEPSOR

Version: 1.2

Prepared by:

Aaron Spicker

Amy Hong

Emily Netemeyer

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Last modified: 04/01/2026

1.0 General Information

1.1 Purpose

The purpose of this Design Specifications Document is to define the detailed design of the proposed medical device intended for the early detection of sepsis and to demonstrate how the design satisfies the requirements established in the Requirements Specifications Document. This document converts user needs, regulatory considerations, engineering constraints and performance expectations into concrete design elements, subsystem architectures, and technical implementation choices.

The document presents the device as an integrated system composed of multiple interacting subsystems, including sensing, signal processing, physical housing, power delivery, software, and user interfacing. For each major design element, the document explains the rationale behind the selected approach and shows how that approach supports one or more associated requirements. In this way, the document establishes a clear link between defined requirements and the corresponding design decisions used to satisfy them.

Finally, this document provides the technical foundation for downstream activities such as verification, validation, risk management, prototype refinement, and requirements traceability. It serves as a reference to ensure that all design decisions remain consistent with the intended function, safety, usability and performance expectations of the device throughout development.

1.2 Scope

This document defines the detailed design specifications of the SEPSOR physiological monitoring system designed for early sepsis detection. It describes the engineering design of the device and its major subsystems, including physical configuration and wearability, physiological measurement and sensing, data interpretation for sepsis risk determination, measurement accuracy, power delivery, electrical safety, alert generation, and user notification behavior. The scope of this document is to establish the technical boundaries of the design and document the specific design choices used to satisfy the requirements established in the Requirements Specifications Document.

The document focuses on the technical implementation of the system and the methods by which functional, safety, usability, and performance requirements are achieved. It highlights the relationship between requirements and design elements by organizing the system into functional areas and showing how each area contributes to overall device operation and intended performance.

This document serves as a technical reference for prototype development, design evaluation, subsystem integration, and future verification and validation activities. In doing so,

it provides the design foundation from which the SEPSOR system can be assessed, refined, and advanced through subsequent stages of development.

1.3 Referenced documents

- 1.3.1 Customer Needs Documents
- 1.3.2 Requirements Specifications Document
- 1.3.3 Requirements Traceability Matrix
- 1.3.4 Design Traceability Matrix
- 1.3.5 ISO 15223-1
- 1.3.6 ELEGOO, MB102 Breadboard Power Supply Module Datasheet
- 1.3.7 ELEGOO, Active Buzzer Datasheet
- 1.3.8 Seeed Studio, XIAO ESP32C3 Datasheet, SKU 113991054
- 1.3.9 STMicroelectronics, LSM6DSO32 Datasheet, DS12814
- 1.3.10 FDA 21 CFR 820
- 1.3.11 ISO 10993
- 1.3.12 IEC 60601-1
- 1.3.13 IEC 60601-1-2
- 1.3.14 IEC 60601-1-8
- 1.3.15 IEEE 1708
- 1.3.16 DHHS (NIOSH) Publication No. 98-126
- 1.3.17 IEC 60529

1.4 Acronyms and Abbreviations

- 1.4.1 ADC: Analog-to-Digital Converter
- 1.4.2 APB: Advanced Peripheral Bus
- 1.4.3 BLE: Bluetooth Low Energy
- 1.4.4 BPM: Beats Per Minute
- 1.4.5 CC-CV: Constant Current-Constant Voltage
- 1.4.6 cm: centimeter
- 1.4.7 DFT: Discrete Fourier Transform

1.4.8 DoF: Degrees of Freedom

1.4.9 EMC: Electromagnetic Compatibility

1.4.10 ESP32-C3: Espressif 32-bit microcontroller, C3 variant

1.4.11 FFT: Fast Fourier Transform

1.4.12 FIFO: First In, First Out

1.4.13 GPIO: General Purpose Input/Output

1.4.14 HR: Heart Rate

1.4.15 Hz: Hertz

1.4.16 I2C: Inter-Integrated Circuit

1.4.17 I²S: Inter-IC Sound

1.4.18 IBI: Inter-Beat Interval

1.4.19 IDFT: Inverse Discrete Fourier Transform

1.4.20 IIR: Infinite Impulse Response

1.4.21 IMU: Inertial Measurement Unit

1.4.22 ISR: Interrupt Service Routine

1.4.23 Li-Po: Lithium-Polymer

1.4.24 LSB: Least Significant Bit

1.4.25 mA: milliampere

1.4.26 mAh: milliamp-hours

1.4.27 MCU: Microcontroller Unit

1.4.28 MEMS: Microelectromechanical Systems

1.4.29 mmHg: Millimeters of Mercury

1.4.30 OVP: Overvoltage Protection

1.4.31 PCM: Protection Circuit Module

1.4.32 PTT: Pulse Transit Time

1.4.33 PWM: Pulse Width Modulation

- 1.4.34 QR: Quick Response
- 1.4.35 RF: Radio Frequency
- 1.4.36 RISC-V: Reduced Instruction Set Computer V
- 1.4.37 RR: Respiratory Rate
- 1.4.38 SCG: Seism cardiography
- 1.4.39 SPI: Serial Peripheral Interface
- 1.4.40 SRAM: Static Random-Access Memory
- 1.4.41 TPU: Thermoplastic Polyurethane
- 1.4.42 UART: Universal Asynchronous Receiver-Transmitter
- 1.4.43 U.FL: Ultra Miniature Coaxial
- 1.4.44 UDI: Unique Device Identifier
- 1.4.45 USB-C: Universal Serial Bus Type-C
- 1.4.46 UVP: Undervoltage
- 1.4.47 Protection
- 1.4.48 Wh: Watt-hour
- 1.4.49 μ A: Microampere

2.0 Overall Description

This device is designed to support the early detection of sepsis through the monitoring of three key physiological markers: HR, RR, and BP. While sepsis is commonly identified and managed in clinical settings, individuals recovering outside of direct medical supervision may face delayed recognition of early warning signs. Accordingly, the device is intended for users at elevated risk of sepsis who are outside of continuous hospital care, including recently discharged patients, post-operative patients, individuals participating in outpatient recovery, and other high-risk populations such as those with underlying medical conditions or recent injuries.

The primary function of the device is to notify users when monitored physiological patterns indicate elevated sepsis risk, with the goal of reducing time to clinical evaluation/intervention for a condition in which early recognition can significantly improve survival and overall health outcomes. To support this purpose, the device is designed for continuous, noninvasive use during normal daily activity and home recovery. The overall design emphasizes

reliability, accessibility, affordability, and ease of use, enabling a monitoring experience that is supportive, practical, and minimally disruptive to the user.

2.1 Assumptions

The following assumptions define the expected user population, operating environment, and system usage conditions for the SEPSOR physiological monitoring system. These assumptions establish the context in which the device is designed to operate and provide the basis for subsystem design, performance expectations, and verification planning. They reflect the intended use case of the device and support the development of a focused, consistent, and technically appropriate design.

2.1.1 User Assumptions

2.1.1.1 The intended user is a patient identified by a licensed healthcare provider as being at risk for developing sepsis.

2.1.1.2 The intended user is eligible to receive physiological monitoring outside of an acute inpatient hospital setting.

2.1.1.3 The intended user has baseline HR values within 60 to 100 BPM.

2.1.1.4 The intended user has baseline RR values within 12 to 18 breaths per minute.

2.1.1.5 The device is intended for use during routine daily activities.

2.1.1.6 The device is intended for use during low intensity physical activities.

2.1.1.7 The intended user is able to hear audible alerts at the defined sound levels.

2.1.1.8 The user is cognitively able to recognize the specified alert sounds.

2.1.1.9 The user is able to respond to the specific alert sounds appropriately.

2.1.1.10 The intended user can independently turn on the device without assistance.

2.1.1.11 The intended user can recharge the device without assistance.

2.1.1.12 The intended user can reposition the device without assistance.

2.1.1.13 The intended user has intact skin at the wear location.

2.1.1.14 The intended user has no known allergies to medical-grade adhesive.

2.1.1.15 The intended user has no known skin sensitivity to medical-grade adhesive.

2.1.1.16 The intended user has no known allergies to the strap materials.

2.1.1.17 The intended user has no known skin sensitivity to the strap materials.

2.1.1.18 The intended user follows the defined charging schedule to maintain continuous monitoring capability.

2.1.2 Environmental Assumptions

2.1.2.1 The device is primarily used in indoor, temperature-controlled environments, representative of a home recovery setting.

2.1.2.2 The ambient operating temperature is between 60-80 °F.

2.1.2.3 The ambient operating humidity levels are between 30 to 60%.

2.1.2.4 The device is exposed to minimal amounts of water from sweat.

2.1.2.5 The device is exposed to minimal amounts of water from normal skin moisture.

2.1.2.6 Electromagnetic interference levels are consistent with typical residential environments.

2.1.2.7 The normal storage conditions are 0-50 °C.

2.1.2.8 The normal storage conditions are in moderate relative humidity.

2.1.3 Operational Assumptions

2.1.3.1 The device is used as an early warning tool for sepsis.

2.1.3.2 The device is used as a screening tool rather than a standalone diagnostic system for sepsis.

2.1.3.3 Sensor-to-skin contact is always maintained during normal use to ensure valid physiological measurements.

2.1.3.4 The device is worn in the intended chest location.

2.1.3.5 The device is worn in the intended orientation.

2.2 Constraints

2.2.1 The device must comply with IEC 60601-1 electrical safety requirements.

2.2.2 The device must comply with IEC 60601-1-2 electromagnetic compatibility requirements.

2.2.3 Audible alerts must comply with IEC 60601-1-8 alarm standards.

2.2.4 All skin-contacting materials must meet ISO 10993 biocompatibility requirements.

2.2.5 Device components must cost no more than \$100.

2.3 Risks

2.3.1 Missed, Delayed, or Incorrect Detection of Sepsis Risk

2.3.1.1 Variability in physiological signals or their interpretation may result in missed, delayed, or incorrect identification of sepsis risk.

2.3.1.2 This risk may lead to unnecessary alerts or delayed/missed intervention opportunities on the part of the user.

2.3.2 Signal Acquisition Degradation and Measurement Accuracy

2.3.2.1 Improper placement, loss of sensor contact, or extreme motion during use may degrade signal acquisition and measurement accuracy.

2.3.2.2 This risk may result in reduced reliability of data and downstream results.

2.3.3 User Awareness and Response Limitations

2.3.3.1 Audible alerts may not be perceived or appropriately acted upon due to environmental or user-dependent factors.

2.3.3.2 This risk may reduce the early warning capabilities of the device.

2.3.4 Power and Operational Continuity

2.3.4.1 Battery depletion or degradation may occur if charging requirements are not upheld.

2.3.4.2 This risk may interrupt continuous monitoring capabilities or interfere with functionality entirely depending on severity.

2.3.5 Electrical Safety

2.3.5.1 Electrical component faults or abnormal operating conditions may result in unintended electrical behavior.

2.3.5.2 Localized heating may occur during operation or charging, particularly in fault conditions.

2.3.5.3 These risks may result in user discomfort or minor injury as well as degradation or device malfunction under fault conditions.

3.0 System Design

3.1 System Architecture

3.1.1 System Components

3.1.1.1 The sensing system consists of two accelerometers that generate motion signals.

3.1.1.2 The processing subsystem consists of a microcontroller that performs signal acquisition as well as conversion and analysis of acquired data.

3.1.1.3 The user feedback subsystem consists of audio output capabilities for system status and risk alerts.

3.1.1.4 The power subsystem consists of a rechargeable battery and regulation circuitry to supply a stable operating voltage to all subsystems.

3.1.1.5 The sensing, processing, feedback, and power subsystems are electrically interconnected to support continuous data flow from system acquisition through user feedback.

3.1.1.6 The device integrates these subsystems into a wearable assembly alongside enclosure and attachment components that allows for consistent operation.

3.1.2 Input/Data Processing

3.1.2.1 Physiological input to the system takes the form of mechanical chest motion signals corresponding to respiration, cardiac activity, and blood flow.

3.1.2.2 Input signals represent localized motion of the chest.

3.1.2.3 Input signals represent spatial variation across the chest.

3.1.2.4 The system converts acquired signals into digital data for processing within the device.

3.1.2.5 The system derives the following physiological parameters from incoming data: ~~BP~~, HR, and RR.

3.1.2.6 The system evaluates the aforementioned parameters as a part of determining a sepsis risk score.

3.1.3 User Feedback

3.1.3.1 The system translates processed physiological data into user-facing alerts when alert conditions are met.

3.1.3.2 Alert generation is triggered by defined physiological conditions and system states.

3.1.3.3 The system differentiates between alert outputs to distinguish risk warnings from battery notifications.

3.2 User Interface

3.2.1 User interfacing occurs in the form of audible alerts that communicate device status and relevant warnings to the user.

3.2.2 The alert system is implemented via a buzzer driven by the device microcontroller using a PWM signal.

3.2.3 Two distinct alert signals are used to differentiate risk warnings from system notifications to the user.

3.2.4 Sepsis alerts are prioritized over routine status alerts.

3.3 Packaging and Labeling

3.3.1 The device is packaged in a sealed, medical-grade pouch that protects it from moisture.

3.3.2 The device remains intact under normal storage conditions.

3.3.3 The device is held in place with a simple internal insert to prevent damage during shipping.

3.3.4 The package is easy to open without tools.

3.3.5 The device label includes a UDI that is readable by scanner.

3.3.6 The label prints all information in a clear, readable font.

3.3.7 Required ISO 15223-1 symbols are printed adjacent to the UDI.

3.3.8 Label printing resists smearing from normal cleaning.

3.3.9 A QR code on the label links users to digital instructions and support resources.

3.3.10 Packaging for SEPSOR



3.4 Instructions for Installation, Use, and Servicing

3.4.1 Instructions include simple diagrams showing where to place the device on the chest.

3.4.2 Instructions include simple diagrams showing how to place the device on the chest.

3.4.3 Instructions describe basic skin preparation.

3.4.4 Instructions warn users about how incorrect placement may affect measurements.

3.4.5 Instructions explain how to turn on the device to confirm it is working.

3.4.6 Instructions clearly describe alert types.

3.4.7 Instructions clearly describe what the user should do for each alert type.

3.4.8 A simple troubleshooting section addresses common issues.

3.4.9 Instructions explain how to safely recharge the device.

3.4.10 Instructions list basic precautions to ensure assumptions are met.

3.4.11 Contact information for support/warranty service is included.

SEPSOR INSTRUCTION MANUAL

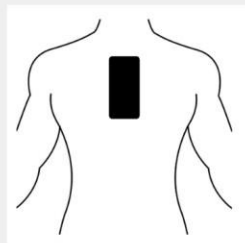
Package Contents

- 1x SEPSOR Sepsis Alert Device
- 1x Adjustable Velcro Chest Strap
- 20x Medical-grade adhesive pads
- 1x USB charging cable
- 1x Instruction manual

SEPSOR is a device that continuously monitors respiratory rate, heart rate, and blood pressure to determine the wearer's risk of determining sepsis. SEPSOR provides alerts to the wearer if abnormal physiological measurements are detected.

How to Wear SEPSOR

1. Prepare the Skin
 - a. Clean the placement area with mild soap and water
 - b. Dry thoroughly
2. Attach the Adhesive Pad
 - a. Align the pad with the back of the device housing
 - b. Press firmly to attach device
3. Position the Device on the Chest
 - a. Place the device on the upper chest, directly under below the collarbones.



4. Secure with the Velcro Strap
 - a. Thread the strap through the side loops and pull to secure

Alerts

1. Sepsis Detection
 - a. Device will emit a loud sequence of loud beeps when the device has detected the wearer is at risk of sepsis
 - b. Alert 911 and seek medical attention after this alert
2. Low Battery
 - a. Device will beep three times in succession to alert the user that the device battery is low
 - b. Remove the device and allow it to recharge after this alert

Maintenance and Servicing

- Avoid exposure to extreme temperatures, submersion in water, or direct sunlight
- Replace chest adhesives regularly
- Do not attempt to open or repair internal components

Important Safety Information

- Do not use on broken or irritated skin
- Do not attempt to open or modify the device housing
- Keep away from excessive heat or moisture
- This device is not a substitute for professional medical diagnosis or treatment
- If you experience dizziness, chest pain, or shortness of breath, seek immediate medical attention

4.0 Component Design

4.1 Hardware Design

4.1.1 Physiological Measurement System

4.1.1.1 The device measures physiological signals using two accelerometers positioned on the chest.

4.1.1.2 One accelerometer is positioned near the xiphoid process to detect motion of the lower sternum and chest wall.

4.1.1.3 A second accelerometer is positioned at the suprasternal notch to detect mechanical vibrations generated by cardiac activity and blood flow in the upper thoracic region.

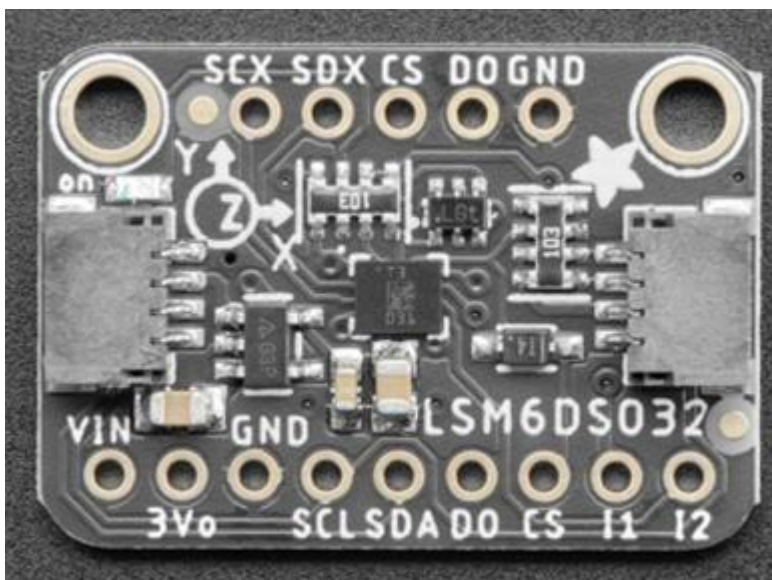
4.1.1.4 The accelerometer located near the xiphoid process also detects periodic chest motion associated with heart rate to enable heart rate measurement.

4.1.1.5 Sensor signals are transmitted to the device processing electronics and microcontroller unit where filtering and signal processing algorithms extract physiological parameters.

4.1.1.6 The accelerometers are positioned 17.5 cm apart vertically for correct anatomical placement.

4.1.1.7 Accelerometer Module Specifications

Top view



Bottom View

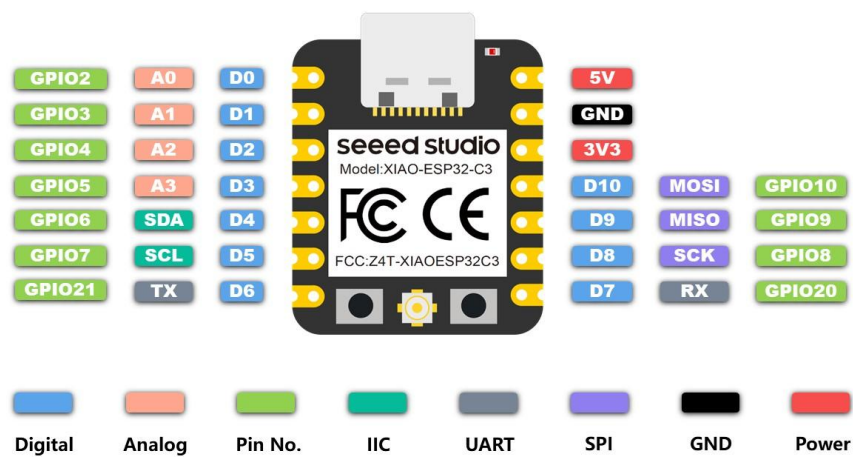


Parameter	Specification
Component	Adafruit LSM6DSO32 IMU (Product 4692)
Manufacturer (Module)	Adafruit Industries LLC
Manufacturer Part Number (Module)	4692
Product Name	Adafruit LSM6DSO32 6-DoF Accelerometer + Gyroscope
Sensor IC Manufacturer	STMicroelectronics
Sensor IC Part Number	LSM6DSO32
Sensor Type	3-axis MEMS accelerometer with integrated 3axis gyroscope
Dimensions	25±1 mm × 17±1 mm × 4±1 mm
Degrees of Freedom	6 DoF
Axes	X, Y, Z

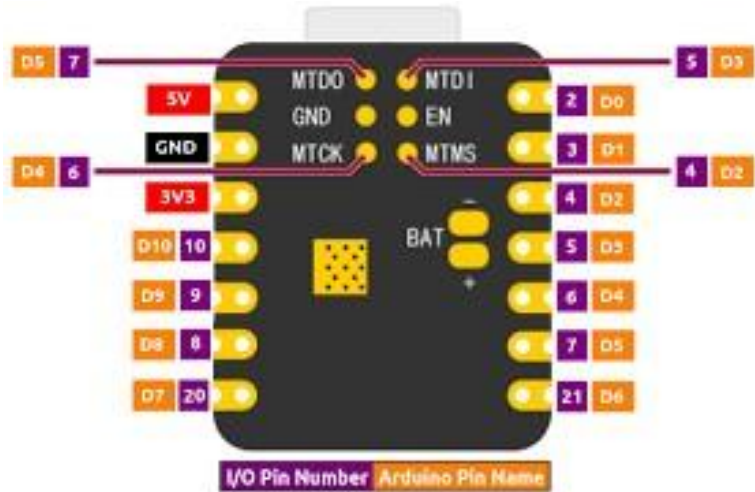
Accelerometer Range	±4 g, ±8 g, ±16 g, ±32 g
Resolution	16-bit (~0.122 mg/LSB at ±4 g)
Output Data Rate	1.6 Hz to 6.7 kHz
Noise Density	~90–120 µg/√Hz
Interface	I2C and SPI
Operating Voltage (Board)	3.0 V – 5.0 V
Operating Voltage (Sensor)	1.71 V – 3.6 V
Current Consumption	~0.55 mA active, <10 µA low power
Temperature Range	–40°C to +85°C
Embedded Features	Tap, free-fall, wake-up, activity detection
FIFO Buffer	~9 kB
Connectivity	STEMMA QT / Qwiic
Programming Support	Arduino, CircuitPython, Python, C/C++
Primary Function	Measures linear acceleration

4.1.1.8 Microcontroller Specifications

Top View



Bottom View



Section	Parameter	Value
Product Information	Manufacturer	Seeed Technology Co., Ltd
	Part Number	113991054

	Product Name	XIAO ESP32-C3
	MCU Architecture	32-bit RISC-V
	Wireless Capability	Wi-Fi + Bluetooth
Microcontroller	Processor	ESP32-C3
	Flash Memory	4 MB
	SRAM	~400 KB
	Clock Speed	160 MHz
Wireless Communication	Wi-Fi Standard	802.11 b/g/n (2.4 GHz)
	Bluetooth Standard	Bluetooth 5.x (BLE)
	RF Frequency	2.4 GHz
Power Specifications	Input Voltage (USB)	5 V
	Operating Voltage	3.3 V
	On Board Battery Charge IC	Yes
Interfaces / Peripherals	GPIO Pins	Multiple programmable
	UART	Supported
	SPI	Supported
	I²C	Supported
	I²S	Supported
	ADC	Supported
	PWM	All GPIO pins are capable

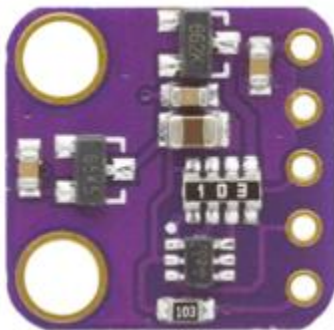
	Timers	- 2 × 54-bit general-purpose timers - 3 × watchdog timers - 1 × 52-bit system timer
Connectivity / Hardware	Antenna Interface	U.FL connector supported
	USB Interface	USB-C (power + programming)
	Headers	Not pre-soldered
Mechanical Characteristics	Dimensions	21 × 17.5 mm
	Board Type	Ultra-compact
RF Characteristics	Wireless Protocols	Wi-Fi 4 + BLE
	RF Function	Integrated transceiver
Operating Features	Development Platform	Arduino / ESP-IDF
	Applications	IoT, embedded systems, wearables

4.1.1.9 Heart Rate & SPO2 Sensor Module

Top View



Bottom View



Parameter	Specification
Component	Pulse Oximeter / Heart Rate Sensor Module (PPG)
Manufacturer (Module)	SunFounder
Manufacturer Part Number (Module)	4411-ST0244-ND
Product Name	MAX30102 Pulse Oximeter and Heart Rate Sensor
Sensor IC Manufacturer	Maxim Integrated
Sensor IC Part Number	MAX30102
Sensor Type	Optical (Photoplethysmography)
Dimensions	14 x 14 x 1.5 mm

Interface	I2C
Operating Voltage (Board)	3.3V or 5V
Operating Voltage (Sensor)	~1.8V
Current Consumption	< 5 mA
Temperature Range	-40 °C to 85 °C
Embedded Features	Onboard red + IR LEDs; integrated signal processing
FIFO Buffer	Yes (integrated)
Connectivity	I2C (SCL, SDA, INT)
Programming Support	Arduino (SparkFun MAX3010x library)
Primary Function	Heart rate (BPM) and blood oxygen saturation (SpO ₂) measurement via optical PPG

4.1.2 Device Enclosure

4.1.2.1 The device electronics and sensors are enclosed in a 3D-printed TPU housing.

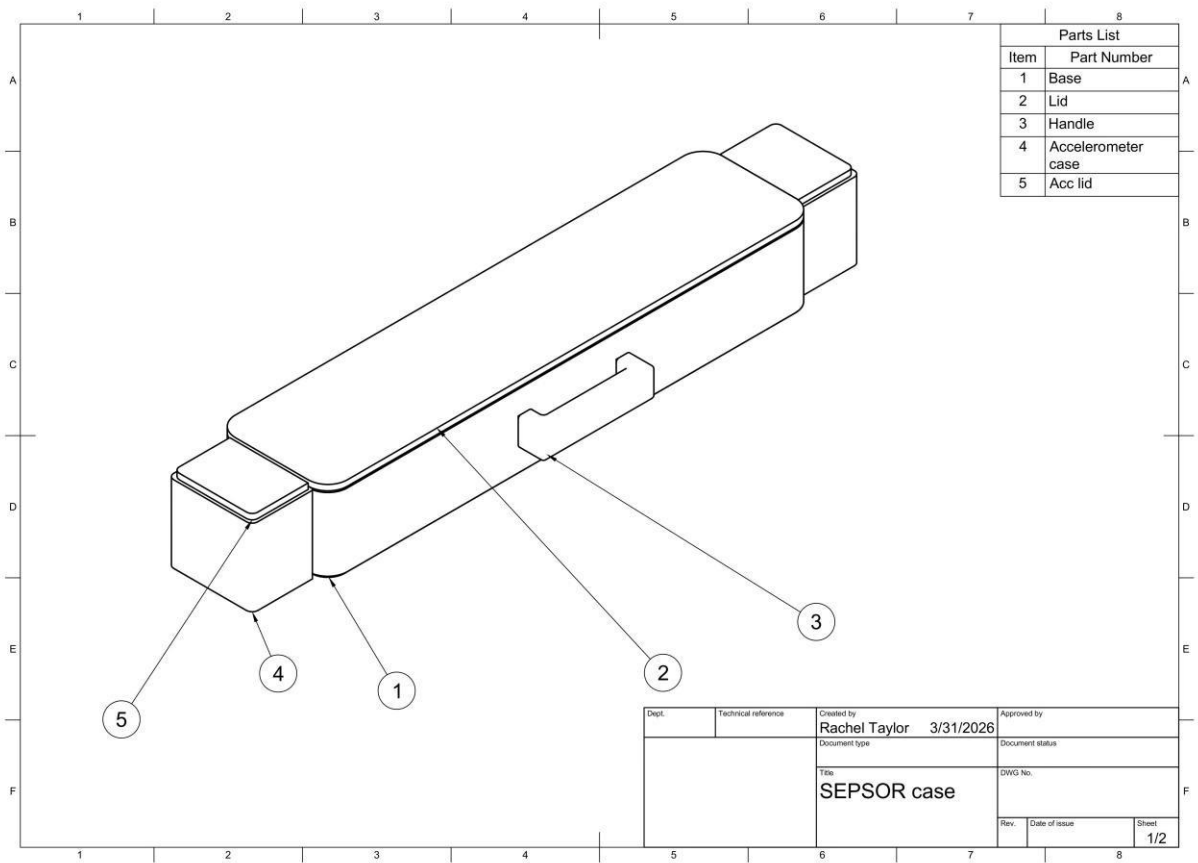
4.1.2.2 The TPU enclosure provides flexibility that allows the device to conform to the curvature of the user's chest.

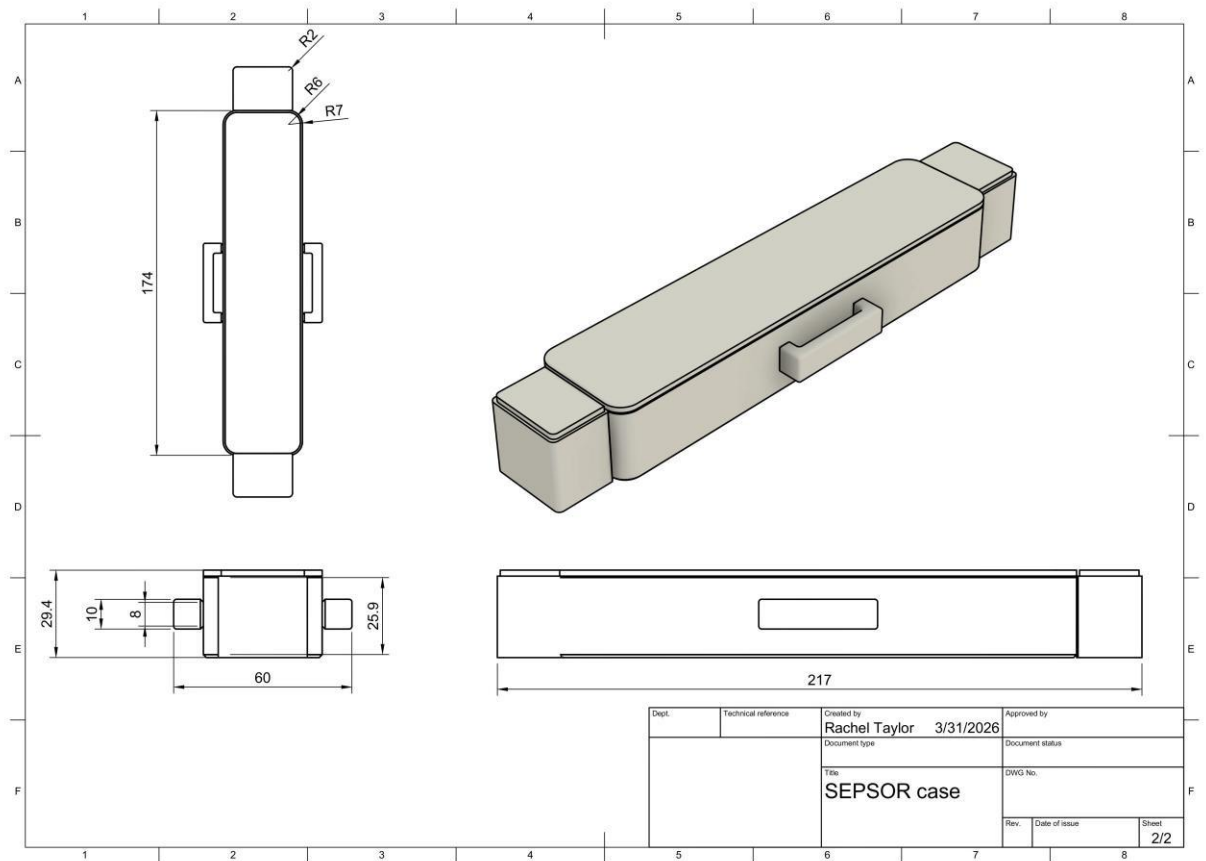
4.1.2.3 The enclosure protects internal components from sweat, moisture, and minor environmental exposure during normal operation.

4.1.2.4 The enclosure contains internal mounting features that secure the sensors, battery, and processing electronics.

4.1.2.5 The enclosure is 21.7 cm x 6 cm x 2.94 cm

4.1.2.6 Enclosure Design Drawing





4.1.3 Device Attachment System

4.1.3.1 The device maintains direct sensor contact with the user's chest during operation.

4.1.3.2 A medical-grade micro foam adhesive pad is located on the back surface of the device to maintain sensor-to-skin contact.

4.1.3.3 The adhesive covers the entire back and extends 1 cm from the edge of the enclosure in all directions.

4.1.3.4 The device includes a Velcro chest strap that wraps around the user's torso to provide additional stabilization.

4.1.3.5 The attachment system prevents displacement of the device during normal daily activities.

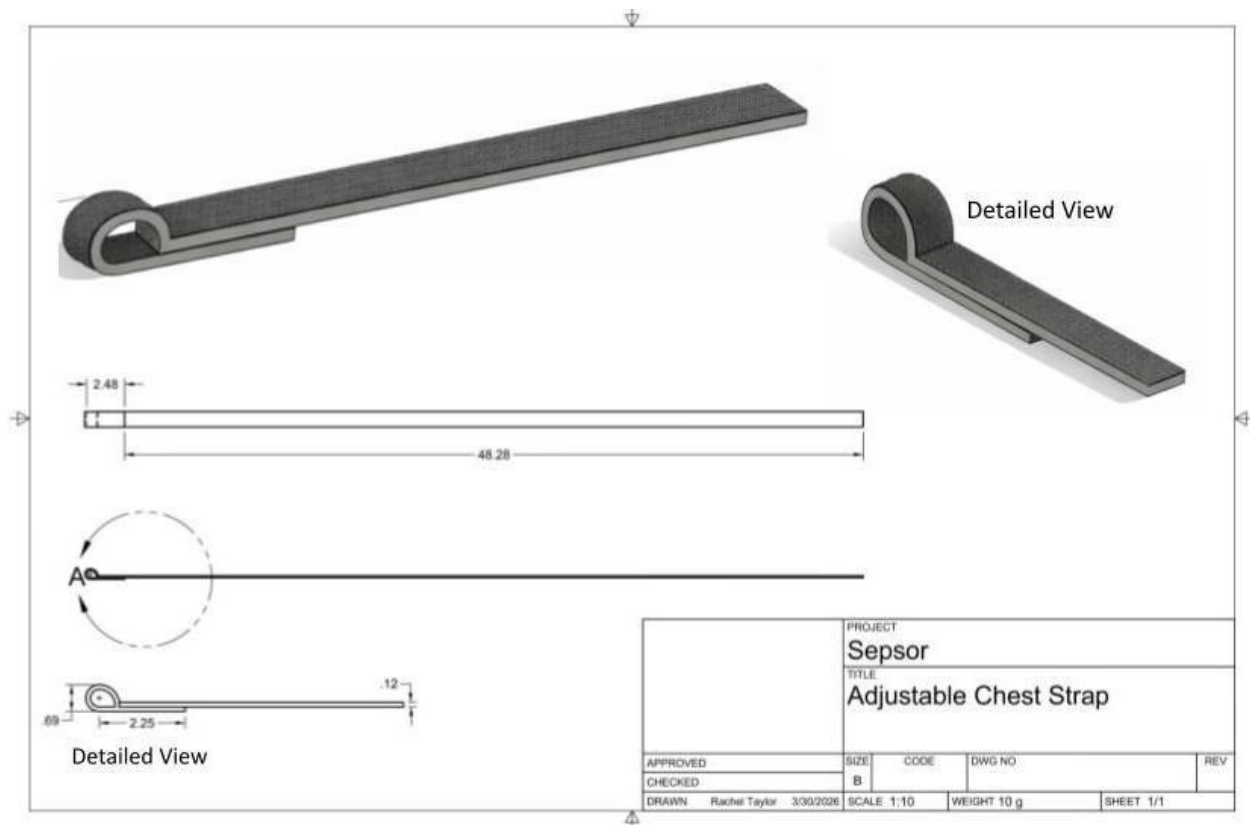
4.1.3.6 Adhesive Specifications

Parameter	Value
Height	1 mm
Length	21 mm
Width	76.2 mm
Item Weight	31.751 g
Material	Foam, Rubber
Brand	3M
Manufacturer	3M
Part Number	15283

4.1.3.7 Strap Specifications

Height	0.12 ± 0.04 in
Length	50.76 in
Width	1.0 in
Item Weight	52 grams
Material	Nylon
Brand	YMCRLUX
Manufacturer	YMT
Part Number	WVS2505

4.1.3.8 Strap Design Drawing



4.1.4 Power System

4.1.4.1 The device will be powered by an internal rechargeable battery.

4.1.4.2 The battery supports continuous physiological monitoring operation.

4.1.4.3 The device includes a USB-C charging interface to allow the battery to be recharged.

4.1.4.4 The power system provides regulated voltage to the sensors and processing electronics.

4.1.4.5 Lithium-Polymer Battery Datasheet

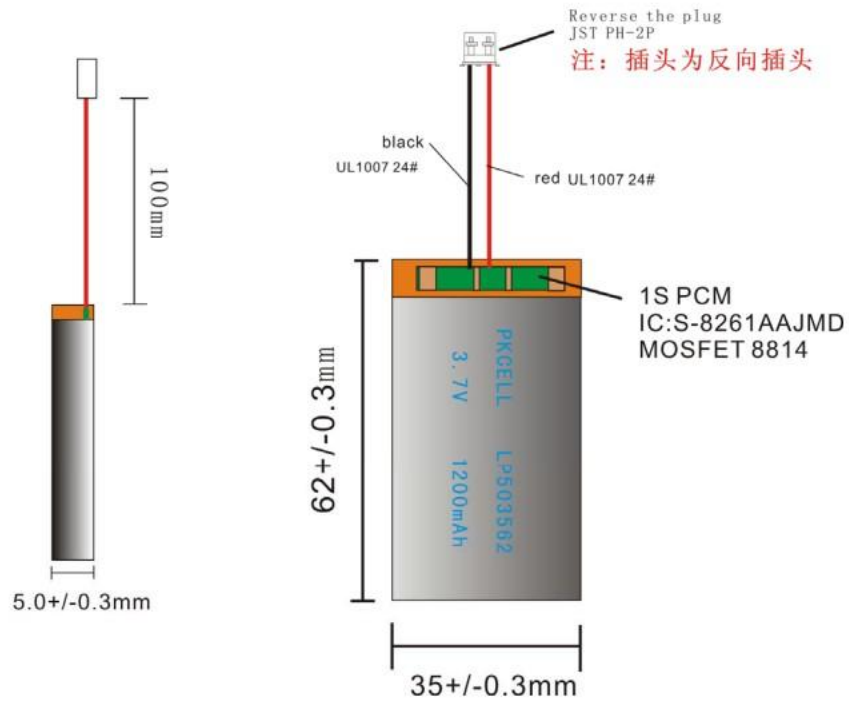
Section	Parameter	Specification	Notes
General	Model	LP503562	Li-Po Battery
	Chemistry	Lithium-Ion Polymer	Rechargeable
	Configuration	1S (single cell)	

	Protection	Integrated PCM	OVP, UVP, SCP
Electrical Characteristics	Nominal Voltage	3.7 V	
	Nominal Capacity	1200 mAh	
	Minimum Capacity	1140 mA	
	Energy	4.4 Wh	
	Maximum Charge Voltage	4.2 V	
	Discharge Cutoff Voltage	3.0 V	
Charging Specifications	Charge Method	CC–CV	Constant Current/Voltage
	Standard Charge Current	0.2C (240 mA)	
	Fast Charge Current	1C (1200 mA)	
	Charge Termination Current	0.01C	

Discharging Specifications	Standard Discharge Current	0.2C	
	Maximum Continuous Discharge Current	1C (1200 mA)	
Environmental Specifications	Charging Temperature	0 to 45 °C	
	Discharging Temperature	–20 to 60 °C	

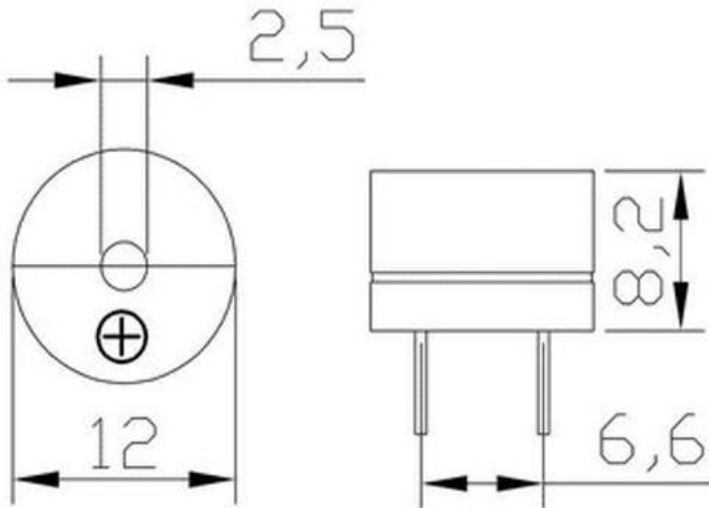
	Storage Temperature (Short-Term)	-20 to 45 °C	
	Storage Temperature (Long-Term)	-10 to 35 °C	
Mechanical Characteristics	Dimensions	5 × 35 × 62 mm	
	Weight	~24 g	
	Form Factor	Pouch Cell	
Performance Characteristics	Cycle Life	>300 cycles	to 60% capacity
	Capacity Retention	≥80%	after 28 days
	Self-Discharge Rate	Low	Typical Li-Po
Protection Features	Overcharge Protection	Yes	PCM
	Over-discharge Protection	Yes	PCM
	Short-circuit Protection	Yes	PCM
Safety Limits	Maximum Voltage Limit	4.2 V	
	Minimum Voltage Limit	3.0 V	
	Maximum Operating Temperature	60 °C	
Design Integration	Operating Voltage Range	3.0 – 4.2 V	
	Recommended Charger	Li-Po CC–CV	4.2 V termination

4.1.4.6 Battery Pack Drawings



4.1.5 Ancillary Hardware

4.1.5.1 Diagram for the buzzer used to indicate sepsis or low battery

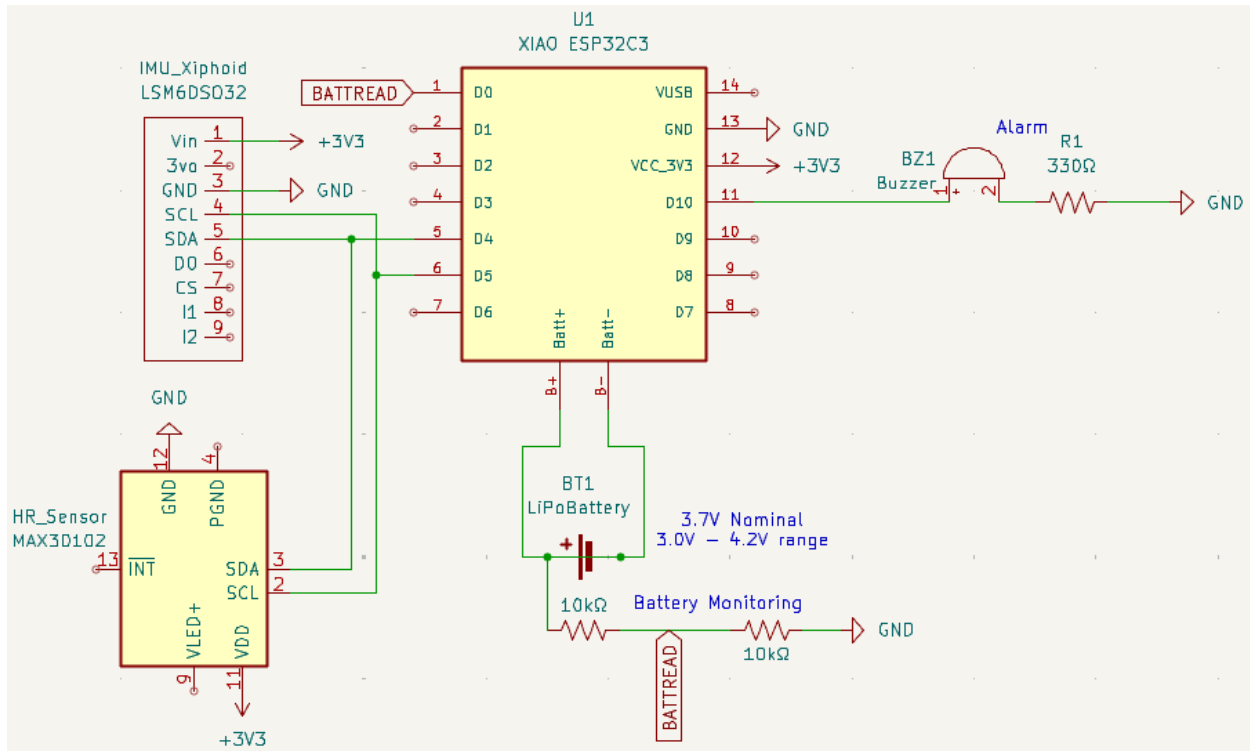


4.1.5.2 Buzzer Specifications

Buzzer Type	Passive
Voltage Rating	5V
Voltage Input Range	3 - 12V
Resistance	16 Ohms
Loudness at 10cm	85dB
Input Signal Type	PWM

4.2 Circuit Design

4.2.1 Device Schematic



As seen in the schematic, the device is powered by a 3.7V LiPo battery connected to the battery + and – pins on the board. The MCU interfaces with an IMU (IMU_Sternum) and a MAX30102 pulse oximeter module over a shared I2C bus. Both devices are connected to the same SCL (D5) and SDA (D4) lines, allowing simultaneous communication as-is, due to each sensor possessing a unique I2C address. The MAX30102 module is powered through the 3.3 V rail, providing heart rate and blood oxygen saturation data complementary to acceleration data from the IMU. The Battery Monitoring Label describes the voltage divider subcircuit that allows the MCU to read battery voltage through Analog Pin 0. The Alarm Label describes the buzzer subcircuit that is used to indicate when a user has sepsis, or the device has low battery. This is driven through PWM using pin D10.

4.2.2 Xiao ESP32-C3 draws 20mA when Wi-Fi and Bluetooth are off.

4.2.3 Alarm subcircuit draws 9.54 mA.

$$I = V / R = 3.3V / (16 + 330) = 9.54 \text{ mA}$$

4.2.4 Battery monitoring subcircuit draws 210 μ A maximum when LiPo is fully charged and provides 4.2V.

$$I = V / R = 4.2V / (20k) = 210 \mu A$$

4.2.5 Each IMU draws 550 μA .

4.2.6 IMU, battery monitoring, and alarm subcircuits draw less than the 200mA operating current of ESP32-C3.

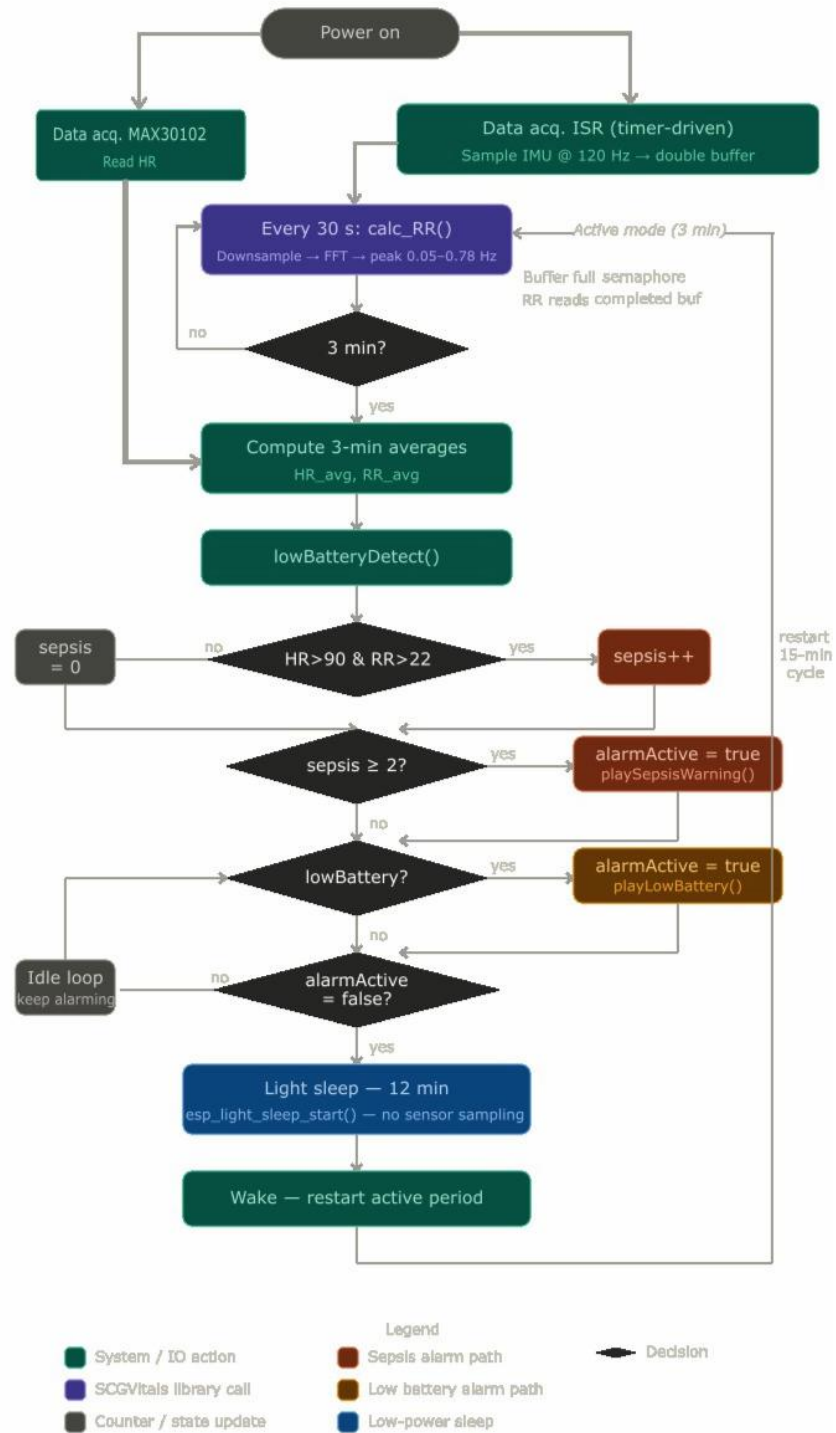
4.2.7 Current Total

Component	Max Current Draw (mA)	Quantity	Subtotal Current Draw (mA)
Xiao ESP32-C3 (Wi-Fi and Bluetooth disabled)	20	1	20
LSM6DSO32 (IMU)	0.55	1	0.55
MAX30102 Module	0.6	1	0.6
Battery Monitoring subcircuit	0.210	1	0.210
Alarm subcircuit	9.54	1	9.54
		Total	30.90

4.2.8 Device duration with 1200 mAh LiPo battery is 38.83 hours

4.3 Software Design

All Software is run on the microcontroller. There are no other devices that are running pertinent software. Below is the overall system design



4.3.1 Overall Implementation

The device software runs entirely on the Seeed Studio XIAO ESP32-C3 microcontroller using the Arduino framework on top of the ESP-IDF runtime. There are no external computing resources; all signal acquisition, digital signal processing, sepsis scoring, alarm management, and power control execute on chip.

The codebase runs entirely in one main script. There are physiological signal processing algorithms for respiratory rate estimation: `calc_RR()` using IMU-derived data, in addition to an internal band-power helper used by `calc_RR()` to determine if the user is wearing the device. HR is acquired directly from the MAX30102 pulse oximeter module via I2C. The main application sketch also contains the hardware initialization, I2C communication between modules, the data-acquisition ISR, the device worn subroutine, the 15-minute duty-cycle state machine, the battery monitor, and the buzzer alarm routines.

Communication between the ISR and the processing layer uses a double-buffer scheme. While the ISR fills one buffer the main loop processes the completed buffer. A volatile semaphore flag, set by the ISR when a buffer is full, signals the main loop to flip its read pointer. RR operates on buffered IMU data, while HR values are read directly over I2C, eliminating the need for trailing window processing in heart rate estimation.

The duty cycle consists of a 3-minute active period followed by a 12-minute light-sleep period, yielding a nominal 15-minute measurement interval. During the active period the ESP32-C3 runs at full clock speed, the IMU ISR fires at 120 Hz from a dedicated hardware timer, and vitals are computed continuously. At the end of the active period the main loop evaluates the sepsis score and battery state. If no alarm is active the device enters ESP light sleep, halting the CPU and IMU sampling while the timer peripheral stays armed for the 12-minute wake. If an alarm is active the device skips sleep entirely and idles in a busy-wait loop, sounding the appropriate alarm until the condition clears.

4.3.2 Data Structures

Two fixed-length arrays form the double buffer. Each buffer holds 3,600 samples per axis of IMU data ($120 \text{ Hz} \times 30 \text{ seconds}$), which is the longest window required by any algorithm. The layout and all associated global states are described below.

Field	Type	Size	Description
-------	------	------	-------------

acc_buf[2][3600]	float32	~28.8kB total	Double buffer. Outer index: active vs. completed buffer. Inner index: z-axis acceleration samples from the IMU (120 Hz over 30 s)
buf_index	volatile int	4 B	Write position within the active buffer (0–3599).
buf_ready[2]	volatile bool[2]	2 B	Per-buffer ready flags indicating whether each buffer has been filled and is ready for processing.
active_buf	volatile int	4 B	Index (0 or 1) of the buffer currently being written.
sepsis_count	uint8_t	1 B	Consecutive-abnormal-reading counter. Incremented when all thresholds are met; reset to zero otherwise.
alarm_active	bool	1 B	True when sepsis_count $\geq$ 2 or lowBatteryDetect() returns 1. Prevents light sleep.
batt_critical_v	float32	4 B	Battery voltage at the moment the low-battery threshold was first crossed in the current alarm episode.
batt_alarm_start_ms	uint32_t	4 B	millis() timestamp of low-battery alarm onset, used for the 2-min-on / 10-min-off duty cycle.

4.3.3 Variable Description

Variable	Type	Scope	Description
----------	------	-------	-------------

acc_buf	float32[2][3600]	Global	Double buffer of raw Z-axis IMU acceleration samples used for respiratory estimation.
buf_index	volatile int	Global	Write index in the active buffer; incremented upon storage of sample
buf_ready	volatile bool[2]	Global	Per-buffer ready flags indicating whether each buffer has been filled and is ready for processing.
active_buf	volatile int	Global	Selects which buffer the ISR is currently writing (0 or 1).
t_start_ms	uint32_t	Global	millis() at start of the current active period.
sepsis_count	uint8_t	Global	Consecutive cycles where all sepsis thresholds were exceeded.
alarm_active	bool	Global	True while any alarm condition is asserted.
rr_running_sum	float32	Global	Accumulated valid RR values during the active period.
rr_sample_count	uint16_t	Global	Count of valid RR samples this active period.
batt_critical_v	float32	Global	Voltage at low-battery threshold crossing for hysteresis.
batt_alarm_start_ms	uint32_t	Global	Timestamp of low-battery alarm onset for duty-cycle enforcement.

4.3.4 Module and Function Descriptions

4.3.4.1 calc_RR()

Signature: void calc_RR(int buf_id)

Purpose: Estimates respiratory rate from a 30-second buffer of vertical-axis IMU acceleration data using downsampling, Hann windowing, FFT spectral analysis, and peak detection in the respiratory band.

Input: buf_id — index of the completed double buffer, containing 3,600 samples of z-axis acceleration acquired at 120 Hz over 30 seconds.

Output: The function computes respiratory rate in breaths/min from the dominant spectral component, printing the result to the serial monitor.

Step 1 — Decimation by factor M = 8:

The 3,600-sample 120 Hz input is downsampled to 450 samples at an effective rate of 15 Hz by retaining every 8th sample. No anti-aliasing filter is applied prior to decimation; this is acceptable because respiratory activity (0.05–0.78 Hz) is orders of magnitude below the aliasing boundary imposed by the new Nyquist frequency (7.5 Hz). The discrete-time relationship is:

$$y[n] = z[M \cdot n] \quad n = 0, 1, \dots, 449 \quad M = 8$$

Step 2 — Hann windowing and zero-padding:

A Hann window is applied to the 450 downsampled samples to reduce leakage

$$w[n] = 0.5(1 - \cos(2\pi n / (N-1)))$$

$$x_w[n] = y[n] * w[n]$$

Where N = 450. The windowed sequence is then zero-padded to 512 points.

Step 3 — FFT magnitude spectrum

A 512 point FFT is computed using an effective sampling rate of 15Hz, with complex output being converted to a magnitude spectrum. The frequency resolution is:

$$\Delta f = 15 / 512 \approx 0.0293 \text{ Hz/bin}$$

Step 4 — Spectral peak identification:

The magnitude spectrum is analyzed over the frequency band of respiration (from 0.05 Hz to 0.78 Hz) in order to identify the dominant respiratory component as:

$$k_peak = \operatorname{argmax} |X[k]|, \text{ for } f_k \in [0.05, 0.78] \text{ Hz}$$

$$f_peak = k_peak * \Delta f$$

Step 5 — RR conversion:

The respiratory frequency is converted to breaths per minute.

$$RR = f_peak * 60$$

Example: Subject breathing at 15 breaths/min, respiratory frequency is 0.25 Hz. With $f_s = 15$ Hz and $FFT_SIZE = 512$, the frequency resolution is $\Delta f = 15/512 \approx 0.0293$ Hz/bin. The nearest bin is $k = \operatorname{round}(0.25 / 0.0293) = 9$, giving $f_peak \approx 0.264$ Hz and $RR \approx 15.8$ breaths/min

Pseudocode: void

```
calc_RR(int
buf_id):

for i in 0:N_DS-1:
    sample = accbuf[buf_id] [i * DS_FACTOR]
    hann = 0.5 * (1 - cos(2πi / (N_DS - 1)))
    vReal[i] = sample * hann

X = FFT(vReal, vImag, FFT_SIZE, DS_RATE_HZ)
|X| = magnitude(X)
```

```

bin_low = ceil(RR_FREQ_LOW / FREQ_RES)
bin_high = floor(RR_FREQ_HIGH / FREQ_RES)

peak_mag = 0
peak_bin = bin_low
for i in bin_low:bin_high:
    if [X[i]] > peak_mag:
        peak_mag = [X[i]]
        peak_bin = i

rr_hz = peak_bin * FREQ_RES
rr_bpm = rr_hz * 60

print rr_pm

```

4.3.4.2 Data Acquisition ISR

Purpose: Reads the IMU at exactly 120 Hz and writes samples into the active double buffer.

Timer: The ISR is attached to ESP32 hardware Timer0 via `timerAttachInterrupt()`. The APB clock (80 MHz) is prescaled to 1 MHz (prescaler = 80) and the alarm period is set to 8,333 ticks, giving an exact 8,333 μ s (1/120 Hz) interrupt interval. Because the hardware timer peripheral generates the interrupt at the precise programmed period, no software timing guard such as a `micros()` comparison is required inside the ISR body; each invocation corresponds to exactly one sample period.

On each ISR invocation, the IMU is read over I2C and the z-axis acceleration value is stored into `accbuf[active_buf][buf_index]`. The index `buf_index` is then incremented. The ISR only writes to the buffer if `buf_ready[active_buf]` is false to better manage buffer overwrites. When `buf_index` reaches 3,600, the main loop marks the buffer as ready, flips `active_buf`, and resets `buf_index` to 0 in order to make the completed buffer available for processing.

Pseudocode:

```

// setup():
timer0 = timerBegin(0, prescaler=80, countUp=true) // 1 MHz tick
timerAttachInterrupt(timer0, &ISR_sample, edge=true)
timerAlarmWrite(timer0, 8333, autoreload=true) // 120 Hz
timerAlarmEnable(timer0)

// ISR body (fires every 8,333 us): void IRAM_ATTR ISR_sample():
read_axes() -> acc[Z]
accbuf[active_buf][buf_index] = acc[Z]
buf_index++
if buf_write_idx == 3600:
    buf_index = 0
    buf_ready[active_buf] = 1
    active_buf ^= 1 // flip double buffer

```

4.3.4.3 lowBatteryDetect()

Signature: int lowBatteryDetect()

Purpose: Monitors battery state-of-charge via the analog voltage divider on pin A0 and applies hysteresis to prevent alarm chatter.

Input: None. Reads ADC pin A0 internally; references globals batt_critical_v and batt_alarm_start_ms.

Output: Returns 1 if a low-battery alarm should be active; returns 0 otherwise.

The ADC reading is scaled to battery voltage using the known resistor-divider ratio. The 20% state-of-charge threshold corresponds to a voltage V_{thresh} derived from the LiPo discharge curve. When the voltage first crosses below V_{thresh} , batt_critical_v and batt_alarm_start_ms are latched and the function returns 1. It continues returning 1 until the voltage recovers above batt_critical_v + 0.1 V, at which point the latch clears. If the voltage subsequently falls again, both latch values are reset for the new episode.

Example: Battery falls to 3.42 V (below threshold): returns 1, batt_critical_v = 3.42 V. Battery recovers to 3.52 V: latch clears, returns 0.

Pseudocode:

```

int lowBatteryDetect():
    v_batt = analogRead(A0) * (V_DIV_RATIO) * (3.3 / 4095)
    // Hysteresis: clear latch if recovered if batt_critical_v > 0 and
    v_batt > batt_critical_v + 0.1:
        batt_critical_v = 0    batt_alarm_start_ms = 0
        return 0
    // Threshold crossing if v_batt <= V_THRESH_20PCT:
    if batt_critical_v == 0: // first crossing in this episode
        batt_critical_v = v_batt    batt_alarm_start_ms = millis()
    return 1
    return 0

```

4.3.4.4 playSepsisWarning()

Signature: void playSepsisWarning()

Purpose: Drives the buzzer on pin D10 to produce a repeating urgent tone pattern alerting the user to a confirmed sepsis risk.

Input: None.

Output: Calls tone(D10, 1000, duration) to produce a 1,000 Hz tone via the Arduino tone() API on pin D10, followed by noTone(D10) and a 500 ms silence. The pattern repeats on each call; the enclosing while (sepsis_count ≥ 2) loop in the main sketch controls repetition and re-evaluates the condition between calls.

Pseudocode:

```

void playSepsisWarning(): tone(D10, 1000,
ON_DURATION_MS) // 1000 Hz on pin D10
    delay(ON_DURATION_MS)
    noTone(D10)
    delay(500)           // 500 ms silence between tones

```

4.3.4.5 playLowBattery()

Signature: void playLowBattery()

Purpose: Drives the buzzer on pin D10 to produce a triple-beep pattern alerting the user to a low battery condition.

Input: None.

Output: Calls `tone(D10, 800, 200)` three times via the Arduino `tone()` API on pin D10, each separated by a 100 ms silence via `noTone(D10)` and `delay(100)`, followed by a 1,500 ms pause before returning. The caller enforces a 2-minutes-on / 10-minutes-off duty cycle using a `millis()` gate before each call.

Pseudocode:

```
void playLowBattery(): for i in {1, 2, 3}:
    tone(D10, 800, 200) // 800 Hz, 200 ms on pin D10
    delay(200)
    noTone(D10)
    delay(100) // 100 ms gap between beeps
    delay(1500) // 1.5 s pause before pattern can repeat
```

4.3.4.6 Main Loop State Machine

Purpose: Orchestrates the 15-minute measurement duty cycle, vital-sign averaging, sepsis scoring, alarm management, and power-mode transitions.

Active period (0–3 minutes):

On each loop iteration the sketch checks whether a new HR reading has been acquired from MAX30102, accumulating valid results into running sums and counters. Whenever `buf_ready` is asserted by the ISR (every 30 s), the completed buffer index is captured, `buf_ready` is cleared, and `calc_RR()` is called for the z-axis of the IMU.

End-of-active-period (`millis()` – `t_start_ms` > 180,000):

Three-minute averages are computed and the sepsis condition is evaluated:

```
HR_avg = hr_running_sum ÷ hr_sample_count
RR_avg = rr_running_sum ÷ rr_sample_count
sepsis_condition = (HR_avg > 90) && (RR_avg > 22)
```

`sepsis_count` is incremented if the condition is true and at least one valid RR sample was collected; otherwise it is reset to zero. `lowBatteryDetect()` is called once to update battery state.

Alarm handling:

If `sepsis_count` ≥ 2 , `alarm_active` is set true and `playSepsisWarning()` is called in a while loop. The sepsis condition is re-evaluated on each pass; once it clears the loop exits. If `lowBatteryDetect()` returns 1, `alarm_active` is set true and `playLowBattery()` is called subject to the `millis()`-based duty-cycle gate. Both conditions are evaluated independently so both alarms may be active simultaneously.

Sleep transition:

If `alarm_active` is false the device configures the wakeup timer for 12 minutes and calls `esp_light_sleep_start()`. On wake, `t_start_ms` is updated, all running sums and counters are cleared, `alarm_active` is cleared, and the active period restarts.

Pseudocode: setup():

```

init_IMU()
init_MAX30102()
configure_timer0_120Hz()
t_start_ms = millis(); last_hr_t = millis()

loop():
  // -- Continuous / periodic HR acquisition from MAX30102 --
  hr = read_HR_MAX30102()
  if hr != -1.0:
    hr_running_sum += hr
    hr_sample_count++

  // -- Every 30 s (buf_ready semaphore): RR -- if buf_ready:
  buf_ready = 0
  rr = calc_RR(completed_buf[Z], 3600)
  if rr1 != -1.0:
    rr_running_sum += rr
    rr_sample_count++

  // -- End of 3-minute active
  period -- if millis() - t_start_ms
  >= 180000:
    HR_avg = hr_running_sum / hr_sample_count

```

```

RR_avg = rr_running_sum / rr_sample_count
lowBatteryDetect()
if HR_avg>90 and RR_avg>22 and rr_sample_count>0:
    sepsis_count++
else:
    sepsis_count = 0

// Sepsis alarm
while sepsis_count >= 2:
    alarm_active = true
    playSepsisWarning()
    re-evaluate sepsis_count

// Low battery alarm (duty-cycle gated)
while lowBatteryDetect():
    if (millis()-batt_alarm_start_ms) % 600000 < 120000:
        alarm_active = true
        playLowBattery()

// Sleep if no alarm
if not alarm_active:
    esp_sleep_enable_timer_wakeup(12 * 60 * 1000000ULL)
    esp_light_sleep_start()

// Reset for next cycle
clear_running_sums_and_counts()
alarm_active = false
t_start_ms = millis()

```

4.3.5 Timing and Synchronization

All IMU data are acquired by a single ISR running on Timer0 at exactly 120 Hz. The IMU is read once per ISR invocation and the z-axis acceleration sample is written into the active double buffer, satisfying timing requirements for respiratory rate acquisition.

The sepsis condition evaluation is a comparison of two scalar averages and executes under 1 ms, satisfying the 50 ms bound. The requirement that the alert be generated within 60 seconds of confirmed abnormal conditions (3.2.1.6) is met due to condition check and alarm call occurring immediately at the end of the 3-minute active period, and playSepsisWarning() producing its first tone within milliseconds of being called.

Design Specification Review Minutes: 04-01

Scope: what will and what won't be in it

- Did not include wont because it's a negative
- Maybe add that it is not a medical diagnosis in the scope
- Document excludes (AI, nonlegal enforcement)

Description:

Assumptions:

- Same as requirements basically
- Operational has weird indentation, 2.1.3.2

Constraints:

- Same as requirements

Risks:

- Some risks not discussed since they are assumptions, good
- Looks good, categories are good

System Design:

- No comments

Component Design:

- No questions or comments
- Picture then table looks great, easy to read
- 4.1.5.3 Resistor Specifications?
 - Do we need anything in this section?

Circuit Design:

- No comments

Software Design:

- Just need to fix formatting to match the rest of the doc

Code Review 04/14

ADD COMMENTS

Alarm code for sound:

- One for buzzer
- One for low battery
- Sepsis Warning
 - o Large alert that keeps playing
- Low battery
 - o 3 beeps with delay 100
 - o Delay too short?
- Recovery Voltage
 - o When to stop alarm when user acknowledges it
- 20% threshold for low battery alarm
- Low battery alert stops when user plugs in
- Voltage divider used
 - o Is that ok for 3.7 lipo battery?
 - o Lipo soldered directly to esp board

Sampling for Accelerometer:

- Set configuration for sampling
- Double buffer system
- Get reading and store into correct buffer
- If buffer has been filled, reset index and switch buffer
- Implement procedure from reference paper to calculate RR from the accelerometer data
- Convert Hz to breaths/minute

Sampling for HR:

- Using a pulse ox

All Code

```
/*
 * sepsorMain.ino
 * script to run sepsor system, which measures respiratory rate and heart rate to
 gauge sepsis risk
 * When sepsis risk has been detected, the device plays a 1000 Hz tone with a
 buzzer to alert the user
 * The code also monitors system battery and starts an alarm when battery is less
 than 20%
 */

#include <Adafruit_LSM6DS032.h>
#include <arduinoFFT.h>
#include <Wire.h>
#include "MAX30105.h"
#include "heartRate.h"
#include "WiFi.h"
#include "esp_bt.h"

// Sampling and Buffer configurations
const uint32_t SAMPLE_RATE_HZ = 120;
const uint32_t SAMPLE_PERIOD_US = 1000000UL / SAMPLE_RATE_HZ; // 8333 µs
const uint32_t WINDOW_SEC = 30;
const uint32_t BUFFER_SIZE = SAMPLE_RATE_HZ * WINDOW_SEC; // 3600 samples

// double buffer for acceleration
float accbuf[2][BUFFER_SIZE];

// indices for double buffer
volatile uint8_t active_buf = 0;
uint8_t read_buf;
volatile int buf_index = 0;
volatile bool buf_ready[2] = {false, false};
volatile uint8_t readHR = 0; // semaphore to let main thread know when to read HR
volatile uint8_t readAcc = 0; // semaphore to let main thread know when to sample
from accelerometer
```

```

// IMU and time initialization
Adafruit_LSM6DS032 imu;
uint8_t rr_count = 0;

// HR sensor
MAX30105 particleSensor;
const uint32_t hr_total_samples = SAMPLE_RATE_HZ * 2; // 240
int hr_count = 0;
// HR polling state – accumulated across loop iterations within each 2s window
unsigned long lastBeat = 0;
long irValue;
unsigned long now;
long delta;
float bpm;
float bpmAccum = 0.0f;
uint8_t beatCount = 0;

// variables for biometrics
float rr_running = 0.0;
float RR_avg;
float hr_running = 0.0;
float HR_avg;

// pin for buzzer to play sepsis or low battery warning
const uint8_t buzzPin = 5; // corresponds to GPIO number on Xiao ESP32-C3, not
the digital pin number

uint8_t lowBattery = 0; // flag for lowBattery
uint8_t sepsis = 0; // sepsis detected counter
uint8_t batteryChecked = 0; // flag for whether battery has been checked during
this active period
unsigned long active_start_time = 0; // initial time that low battery buzzer goes
off

// non-blocking low battery buzzer state machine
enum BuzzState { BUZZ_IDLE, BUZZ_ON, BUZZ_GAP, BUZZ_REST };
BuzzState buzzState = BUZZ_IDLE;
uint8_t buzzBeepCount = 0;
unsigned long buzzPhaseStart = 0;

hw_timer_t *timer0 = NULL;

```

```

// ISR to sample RR, switch to other buffer if current buffer is filled, and set
semaphore to readHR
void IRAM_ATTR sampleAcc() {
    // set flag to get HR reading every 2 seconds
    if(!((buf_index + 1) % hr_total_samples)) {
        readHR = 1;
    }
    // read accelerometer sample when timer hits 0, and store z component in the
buffer only if the buffer is ready
    if (!buf_ready[active_buf]) {
        readAcc = 1;
    }
    // check if active buffer is full
    if (buf_index >= (int)BUFFER_SIZE) {
        buf_ready[active_buf] = true;
        active_buf = 1 - active_buf; // switch buffer
        buf_index = 0;
    }
}

void setup() {
    //Serial.begin(115200);
    //while (!Serial);

    // disable WiFi
    WiFi.mode(WIFI_OFF);
    // Disable Bluetooth
    btStop();
    esp_bt_controller_disable();

    // ensure that IMU works with I2C
    if (!imu.begin_I2C(0x6A)) {
        //Serial.println("ERROR: LSM6DS032 not found at 0x6A – check wiring.");
        while (1);
    }

    // ensure that HR module works with I2C
    if (!particleSensor.begin(Wire, I2C_SPEED_FAST)) {
        //UsedefaultI2Cport, 400kHz speed
        //Serial.println("MAX30102 was not found. Pleasecheckwiring/power. ");
    }
}

```

```

    while (1) ; //Infinite loop to stop the program
}

// accelerometer configuration
imu.setAccelRange(LSM6DS032_ACCEL_RANGE_4_G);
imu.setAccelDataRate(LSM6DS_RATE_208_HZ);

// hr sensor configuration
particleSensor.setup(60, 4, 2, 100, 411, 4096);
particleSensor.setPulseAmplitudeRed(0x24);
particleSensor.setPulseAmplitudeIR(0x24);

// configure buzzer pin as digital output
pinMode(buzzPin, OUTPUT);

// 1 MHz base frequency (1 µs resolution)
timer0 = timerBegin(1000000);

// Attach ISR function
timerAttachInterrupt(timer0, &sampleAcc);

// Alarm every 8333 µs = ~120.0048 Hz; autoreload, indefinite repetition
timerAlarm(timer0, SAMPLE_PERIOD_US, true, 0);
}

void loop() {
    // sample from accelerometer
    if (readAcc) {
        readAcc = 0;
        sensors_event_t accel, gyro, temp;
        imu.getEvent(&accel, &gyro, &temp);
        accbuf[active_buf][buf_index] = accel.acceleration.z;
        buf_index++;
    }
    // process the buffer that has been fully written to
    read_buf = 1 - active_buf;
    if (buf_ready[read_buf]) {
        buf_ready[read_buf] = false;
        rr_running += calc_RR(read_buf);
        rr_count++;
    }
}

```

```

// poll HR sensor on every iteration with single fast, non-blocking I2C read
irValue = particleSensor.getIR();
if (checkForBeat(irValue)) {
    now = millis();
    if (lastBeat != 0) {
        delta = now - lastBeat;
        bpm = 60000.0 / (float)delta;
        if (bpm > 20.0 && bpm < 255.0) {
            bpmAccum += bpm;
            beatCount++;
        }
    }
    lastBeat = now;
}
// every 2s commit window average to running total and reset for next window
if (readHR) {
    readHR = 0;
    if (beatCount > 0) {
        hr_running += bpmAccum / beatCount;
        hr_count++;
    }
    bpmAccum = 0.0;
    beatCount = 0;
    lastBeat = 0; // reset so first beat of next window is used
}
// check for low battery once at the start of each active period
if(!batteryChecked) {
    lowBattery = lowBatteryDetect(); // checks if battery is below 20% and isn't
recovering
    batteryChecked = 1; // set flag
    active_start_time = millis(); // record start of active cycle
}
// play lowBattery sound for 2 minutes every 15 minutes to not annoy user
if (lowBattery && (millis() - active_start_time) < 120000) {
    updateLowBattery();
} else if (buzzState != BUZZ_IDLE) {
    noTone(buzzPin);
    buzzState = BUZZ_IDLE;
}
// after ~3 minutes
if (rr_count >= 6) {

```

```

    RR_avg = rr_count > 0 ? rr_running / rr_count : 0.0; // average RR over the 3
minutes
    HR_avg = hr_count > 0 ? hr_running / hr_count : 0.0; // average HR over the 3
minutes
    if(RR_avg > 22 && HR_avg > 90) { // sepsis risk based on RR and HR
        sepsis++;
    }
    else { // reset if no consecutive detections
        sepsis = 0;
    }
    if(sepsis >= 2) { // play sepsis warning indefinitely if sepsis detected
        while(1) {
            now = millis();
            while((millis() - now) < 180000){ // warning played for 3 minutes
                playSepsisWarning();
            }
            esp_sleep_enable_timer_wakeup(420000000); // sleep for 7 minutes
            esp_light_sleep_start();
        }
    }
    // sleep for 12 minutes (720000000 us) if there is no sepsis alarm
    esp_sleep_enable_timer_wakeup(720000000);
    esp_light_sleep_start();
    wake_state_reset();
}

// clear or reset all variables after MCU wakes up again
void wake_state_reset() {
    active_buf = 0;
    buf_index = 0;
    buf_ready[0] = false;
    buf_ready[1] = false;
    readAcc = 0;
    readHR = 0;
    rr_running = 0.0;
    hr_running = 0.0;
    rr_count = 0;
    hr_count = 0;
    bpmAccum = 0.0;
}

```

```

    beatCount = 0;
    lastBeat = 0;
    buzzState = BUZZ_IDLE;
    buzzBeepCount = 0;
    batteryChecked = 0;
}

// FFT configuration
const int DS_FACTOR = 8;           // downsample 120 Hz to 15 Hz
const int N_DS = BUFFER_SIZE / DS_FACTOR; // 450 downsampled points
const int FFT_SIZE = 512;         // next power-of-two  $\geq$  N_DS (zero-pad)
const float DS_RATE_HZ = (float)SAMPLE_RATE_HZ / DS_FACTOR; // 15.0 Hz
const float FREQ_RES = DS_RATE_HZ / FFT_SIZE; // Hz per bin  $\approx$  0.02930 Hz

const float RR_FREQ_LOW = 0.05;    // lower bound frequency in Hz (~3
breaths/min)
const float RR_FREQ_HIGH = 0.78;   // upper bound frequency in Hz (~47
breaths/min)

// FFT working arrays – static so they live in heap, not the ISR stack
double vReal[FFT_SIZE];
double vImag[FFT_SIZE];

// calc_RR function to compute respiratory rate from a completed 30-second z-axis
buffer.
// Steps:
// 1. Downsample from 120 Hz to 15 Hz (keep every DS_FACTOR-th sample)
// 2. Apply Hann window to the N_DS downsampled points
// 3. Zero-pad to FFT_SIZE = 512
// 4. Run FFT and get magnitude
// 5. Find the peak bin in [0.05, 0.78] Hz
// 6. Convert peak frequency to breaths per minute

float calc_RR(int buf_id) {
    // Downsample and Hann window
    for (int i = 0; i < FFT_SIZE; i++) {
        vReal[i] = 0.0;
        vImag[i] = 0.0;
    }
}

```

```

for (int i = 0; i < N_DS; i++) {
    float sample = accbuf[buf_id][i * DS_FACTOR];
    float hann = 0.5 * (1.0 - cosf(TWO_PI * i / (N_DS - 1)));
    vReal[i] = (double)(sample * hann);
}

// Compute FFT magnitudes
ArduinoFFT<double> fft(vReal, vImag, FFT_SIZE, DS_RATE_HZ);
fft.compute(FFTDirection::Forward);
fft.complexToMagnitude(); // vReal[i] has magnitude at bin i

// Find peak in respiratory frequency band
int bin_low = (int)ceilf(RR_FREQ_LOW / FREQ_RES); // bin 2
int bin_high = (int)floorf(RR_FREQ_HIGH / FREQ_RES); // bin 26

// Bandpower check to see if user is wearing device by integrating power
// spectrum over band using trapezoid rule
double bandpower = 0.0;
for (int i = bin_low; i < bin_high; i++) {
    double p0 = vReal[i] * vReal[i];
    double p1 = vReal[i + 1] * vReal[i + 1];
    bandpower += 0.5 * (p0 + p1) * FREQ_RES;
}
if (bandpower < 1.0) return 0.0f;

// identify max frequency
double peak_mag = 0.0;
int peak_bin = bin_low;
for (int i = bin_low; i <= bin_high; i++) {
    if (vReal[i] > peak_mag) {
        peak_mag = vReal[i];
        peak_bin = i;
    }
}

// Convert from Hz to breaths per minute
float rr_hz = peak_bin * FREQ_RES;
float rr_bpm = rr_hz * 60.0;

return rr_bpm;
}

```

```

// function to play 1000Hz sepsis warning with 66% duty cycle repeated over 1.5s
void playSepsisWarning() {
    tone(buzzPin, 1000);
    delay(1000);
    noTone(buzzPin);
    delay(500);
}

// non-blocking low battery buzzer alarm that advances FSM state each call
void updateLowBattery() {
    unsigned long curr = millis();
    switch (buzzState) {
        case BUZZ_IDLE: // start 800 Hz tone
            tone(buzzPin, 800);
            buzzBeepCount = 1;
            buzzPhaseStart = curr;
            buzzState = BUZZ_ON;
            break;
        case BUZZ_ON: // stop tone after playing for 100 ms
            if (curr - buzzPhaseStart >= 100) {
                noTone(buzzPin);
                buzzPhaseStart = curr;
                buzzState = BUZZ_GAP;
            }
            break;
        case BUZZ_GAP: // play 800 Hz tone again if we haven't had 3 beeps, otherwise
stop
            if (curr - buzzPhaseStart >= 100) {
                if (buzzBeepCount < 3) {
                    tone(buzzPin, 800);
                    buzzBeepCount++;
                    buzzPhaseStart = curr;
                    buzzState = BUZZ_ON;
                } else {
                    buzzPhaseStart = curr;
                    buzzState = BUZZ_REST;
                }
            }
            break;
    }
}

```

```

    case BUZZ_REST: // stop tone for 1500ms
        if (curr - buzzPhaseStart >= 1500) {
            buzzState = BUZZ_IDLE;
        }
        break;
    }
}

// analog pin connected to battery + terminal
#define BATTERY A0
// lipo range is 3.1V to 4.2V, but voltage is halved due to voltage divider to
ensure safe input to analog pin
#define MAX_V 2100
#define MIN_V 1550
#define RECOVERY_V 100 // stop alarm when battery has recovered by 0.1 V to avoid
annoying user

uint32_t val; // current voltage val from pin connected to battery + terminal
uint32_t warning_val = 0; // voltage val corresponding to initial low battery
reading
int v_threshold = MIN_V + 0.2 * (MAX_V - MIN_V);

// function that returns 1 if battery is low and not being charged and 0
otherwise
int lowBatteryDetect(){
    val = 0;
    for(uint8_t i = 0; i < 5; i++){
        val += analogReadMilliVolts(BATTERY); // read battery voltage
    }
    val /= 5;
    if(val >= v_threshold){ // compare to threshold corresponding to 20% battery
        warning_val = 0;
        return 0;
    }
    if(!warning_val){
        warning_val = val; // note voltage val of warning
        return 1;
    }
}

```

```

    if(val > warning_val && (val - warning_val) > RECOVERY_V){ // return 0 if
battery is being charged
        return 0;
    }
    return 1;
}

```

Bill of Materials

Item #	Vendor	Part Number	Quantity	Description	Price per unit	Item cost	Who Bought
2	ADAFRUIT	LSM6DSO32	1528-4692-ND	1 6 DOF IMU	\$12.50	\$12.5	Jay
3	REXQualis		RQ-ECK-001	1 Passive Buzzer	\$1	\$1	Jay
4	YMCRLUX		WVS2505	1 Velcro Strap	\$10	\$4	Emily
5	Vitality Medical		15281	1 adhesive tape	\$11	\$4	Rachel
6	Xiao ESP32-C3		1597-113991054-ND	1 MCU	\$5.99	\$4.99	Jay
7	Sunfounder	MAX30102	4411-ST0244-ND	1 Heart Rate Sensor	\$14.82	\$14.82	Jay
8	TPU Printing		---	1 printing at TIW	\$8	\$8	Rachel
9	3.7V Lipo Battery			1 Lipo battery to power device	\$9	\$9	Jay
Project cost						\$58.30	

Supplemental Data

No supplemental data was used for this project.

References

- [1] Code of Federal Regulations, 21 CFR Part 820 — Quality System Regulation, Electronic Code of Federal Regulations (eCFR), Title 21, Chapter I, Subchapter H, Part 820. Available: <https://www.ecfr.gov/current/title-21/chapter-I/subchapter-H/part-820>. Accessed: Feb. 22, 2026.
- [2] International Electrotechnical Commission, IEC 60601-1-8:2006 — Medical Electrical Equipment — Part 1-8: General Requirements for Basic Safety and Essential Performance — Collateral Standard: General Requirements, Tests and Guidance for Alarm Systems. Available: <https://www.iso.org/standard/41986.html>. Accessed: Feb. 22, 2026.
- [3] International Electrotechnical Commission, IEC 60601-1-2:2014+AMD1:2020 — Medical Electrical Equipment — Part 1-2: General Requirements for Basic Safety and Essential Performance — Collateral Standard: Electromagnetic Disturbances — Requirements and Tests. Available: <https://webstore.iec.ch/en/publication/2590>. Accessed: Feb. 22, 2026.
- [4] Institute of Electrical and Electronics Engineers, IEEE Std 1708-2014 — Standard for Wearable, Cuffless Blood Pressure Measuring Devices, Aug. 26, 2014. Available: <https://standards.ieee.org/standard/1708-2014.html>. Accessed: Feb. 22, 2026.
- [5] National Institute for Occupational Safety and Health, Criteria for a Recommended Standard: Occupational Noise Exposure. U.S. Department of Health and Human Services, Centers for Disease Control and Prevention, June 1998. Available: <https://www.nonoise.org/hearing/criteria/criteria.htm>. Accessed: Feb. 22, 2026.
- [6] International Organization for Standardization, ISO 10993-10:2021 — Biological Evaluation of Medical Devices — Part 10: Tests for Irritation and Skin Sensitization, 2021. Available: <https://www.iso.org/standard/75279.html>. Accessed: Feb. 22, 2026.
- [7] International Electrotechnical Commission, IEC 60529 — Degrees of Protection Provided by Enclosures (IP Code). Available: <https://webstore.iec.ch/en/publication/2452>. Accessed: Feb. 22, 2026.
- [8] Cleveland Clinic, “Vital Signs,” Mar. 15, 2023. Available: <https://my.clevelandclinic.org/health/articles/10881-vital-signs>. Accessed: Feb. 23, 2026.

- [9] ASHRAE, Humidity Control Resources, 2025. Available: <https://www.ashrae.org/technical-resources/bookstore/humidity-control-resources>. Accessed: Feb. 23, 2026.
- [10] STMicroelectronics, LSM6DSO32 Datasheet. Available: <https://www.st.com/resource/en/datasheet/lsm6dso32.pdf>. Accessed: Mar. 25, 2026.
- [11] Kaili Electronic, Piezo Buzzer Product Information. Available: https://www.kailitech.net/products_details/1.html. Accessed: Mar. 23, 2026.
- [12] Seeed Studio Forum, “XIAO ESP32C3 Pin Mapping in MicroPython.” Available: <https://forum.seeedstudio.com/t/xiao-esp32c3-pin-mapping-in-micropython/275215>. Accessed: Mar. 25, 2026.
- [13] arXiv, “Article 2407.21433v1.” Available: <https://arxiv.org/html/2407.21433v1#bib.bib38>. Accessed: Mar. 25, 2026.
- [14] Adafruit Industries, Li-Polymer 503562 1200mAh 3.7V with PCM Datasheet. Available: https://cdn-shop.adafruit.com/product-files/258/C101-_Li-Polymer_503562_1200mAh_3.7V_with_PCM_APPROVED_8.18.pdf (cdn-shop.adafruit.com in Bing). Accessed: Mar. 25, 2026.
- [15] Seeed Studio, XIAO ESP32C3 Datasheet, SKU 113991054. Available: <https://mm.digikey.com/Volume0/opasdata/d220001/medias/docus/6209/Seeed%20113991054.pdf>. Accessed: Mar. 25, 2026.
- [16] RexQualis, “REXQualis – North America,” RexQualis Download Page. Available: <https://www.rexqualis.com/download/>. Accessed: Apr. 1, 2026.